SATA 1D and SATA 2D

Review of the Implementations against the Reference Codes, Verification and Geometry Sweeps

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Abstract

The two SATA implementations of the project — `sata1d` (one azimuth line per channel) and `sata2d` (the range-compressed channel cube, one line per range bin, in front of the two-step range-Doppler reconstruction) — are re-read element by element against the reference codes (`SATA1D.txt`: the Python port of `sata_for_topography` and of the DEM version; `sata.txt`: Pau Prats' IDL `sata`, Marc Jaeger's `sata1_5d` and `sata2d`) and against the papers they come from. Every element of the reference is either reproduced, replaced by an equivalent that is tested here, or deliberately left out with the size of the neglected term given. Three things came out of the review: the sub-aperture bookkeeping was compared with the reference cross-fade (no material difference, table in Section 4.1); the Doppler-dependent range lookup of the reference (the `norcmc` branch, which reads the DEM at the range each Doppler bin actually comes from) was added to `sata2d`; and the floor that the first 2-D sweep showed for flat targets was traced to the ported signal generator — a 5.7-sample chirp with hard edges, which makes a channel with a sub-sample bistatic delay something other than a delayed copy of the reference — and removed by generating the pulse band-limited with the exact delay (Section 4.6). Both suites of checks pass (19 for SATA 1D, 15 for SATA 2D). The same 259-case geometry sweep — nine height-profile families of iso-range targets 200 m apart, plus the cross-track ramps — is then run with both algorithms, and every result is shown next to the geometry it was measured on, in 2D and in 3D; for SATA 2D each selected case also carries the 2-D impulse response and the spectral phase error along range and azimuth frequency, in the layout of Fig. 4.20 of Sakar's dissertation.

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1 Scope and sources

The question of this document is whether the two SATA implementations do what the reference algorithm does, and if not, whether the difference is justified and measured. The material re-read for it:

- `SATA1D.txt` — the Python port of the reference used by the group: `sata_for_topography`, called with `(data, delta_C0_array, rref, prf, Nsb, v, squint, wl, r, inverse, sata_osf)`, which corrects one azimuth line with a map $\Delta C_0[x]$ supplied from outside, and `sata(data, pos, ref, dem, demsmoothed, ...)`, the port of the DEM version;
- `sata.txt` — the IDL originals: `sata` (P. Prats: 2-D data, external DEM, the `norcmc` branch for data before range cell migration correction), `sata1_5d` (M. Jaeger: 1D SATA on range sub-looks, so the wavelength of each sub-look is used) and `sata2d` (M. Jaeger: the Doppler axis as a function of range frequency, range blocks with margins, residual range shifts);
- Prats, Reigber, Mallorquí, *Topography-dependent motion compensation for repeat-pass interferometric SAR systems*, GRSL 2005 — the principle (STFT blocks, Doppler → angle → azimuth position, correction in the range-Doppler domain, linear weighting of the overlap);
- Sakar's dissertation, Chapters 4 and 5 — the two-step reconstruction, the diagnostics of Figs. 4.10–4.12, 4.20 and 5.2 (2-D IRF contour, azimuth response, spectral phase error along f_r and f_a), and the topography-dependent model mismatch of a cross-track baseline (Section 5.2);
- the project notes on the earlier findings (the three sub-band bugs, the C0 map check, the C1/C2 and co-registration study, the case matrix), which fix the conventions used below.

The implementations under review are the ones in the repository `dlr-project`: the package `sata1d_code/sata1d`, the one verified in *SATA 1D: How It Works, Verification and Results*, and `sar_reconstruction/ sata2d`, updated on 25 September with the footprint map, and again here with the range lookup of Section 4.4 and the band-limited signal generator of Section 4.6. Notation follows the 1D report: $\mathrm{PRF}_{\mathrm{ch}} = \mathrm{PRF}/N_{rx} = 500$ Hz is the rate of one channel line, $\Delta C_{0,i}(p)$ the residual of scatterer p in channel i, x_t the azimuth cell of the scatterer on the channel line, h_w the footprint half-width, X the cell shift of one $\mathrm{PRF}_{\mathrm{ch}}$ fold.

2 The reference algorithm

2.1 Principle and the sub-aperture length

A target's echo, after range compression, is a chirp in azimuth; inside a short block of T seconds it is close to a tone whose Doppler f gives the angle β under which the platform sees it, $f = (2v/\lambda)\sin\beta$, and the angle gives the position of the target relative to the centre x_c of the block, $x = x_c + r\tan\beta$ (Prats 2005, Eqs. 4–5). Every Doppler bin of the block's FFT therefore points at a piece of ground, and can be multiplied by the phase correction that piece needs. Two blurs compete when choosing T: the platform moves vT during the block, so the position of the origin x_c is uncertain by $\pm vT/2$; and the FFT resolves $1/T$ in Doppler, which through the ruler is $\lambda r/(2vT)$ in position. They are equal at

$$vT = \frac{\lambda r}{2vT} \quad\Longrightarrow\quad \delta x = vT = \sqrt{\frac{\lambda r}{2}} = 309 \text{ m } (r = 766 \text{ km},\ \lambda = 0.25 \text{ m}), \tag{1}$$

which is the "`deltax = sqrt(wl*rref/2)` — best resolution of SATA" of every reference file, and the sub-aperture is $T_{\mathrm{sub,eff}} = 2\lfloor \delta x\,\mathrm{prf}/(2v)\rceil$ samples of the line handed in (= 20 at

$\mathrm{PRF_{ch}}$ = 500 Hz). This is the resolution with which the topography is accommodated: two scatterers closer than δx share one correction.

2.2 The reference kernel, step by step

Table 1 lists what `sata_for_topography` does, in the order it does it, with the formula as written in the file. The IDL `sata` is the same with two differences noted in the last rows: the FFT block holds N_{zp} *real* samples around the sub-aperture (the "margin") instead of zeros, and the correction is read from a DEM at (azimuth position, range) per bin instead of from a 1D map.

step	reference (`SATA1D.txt`)	remark
sizing	`Tsubeff = round(deltax*prf/v*0.5/Nsb)*2`, `Nzp = 2^ceil(log2 Tsubeff)*sata_osf`, hop `Tsub = Tsubeff/2`	the `/Nsb` assumes data at prf/N_{sb}; called with prf = $\mathrm{PRF_{ch}}$ and $N_{sb} = N_{rx}$ it divides twice (bug 1 of the earlier note)
axis	`fsub = Nsb*(arange(Nzp)*prf/Nzp/Nsb - prf/2/Nsb) + dfc; fsub[0] = prf/2 + dfc; roll by Nzp/2 + pfc`	the leading `Nsb` cancels (bug 2); for squint 0 it is `fftfreq` except that the Nyquist bin is labelled +prf/2
ruler	`azpos = r*(tan(betasub) - tan(squint))/v*prf`, `betasub = asin(wl*fsub/2v)`	Eq. 5 of Prats 2005; the `- tan(squint)` presumes an image processed at that squint (bug 3 for the sub-band use)
block	`aux = data[m*Tsub:(m+2)*Tsub]`, zero-padded to `Nzp`, FFT	rectangular block, no analysis window
lookup	`posaux = int(azpos + posix)`, `posix = (m+1)*Tsub`	block centre plus ruler; truncation, not rounding
correction	`ph = -2*pi/wl*delta_C0_array[posaux]`; `aux *= exp(-j*ph)` if inverse	one complex rotation per bin
overlap	`aux[:Tsub] = temp[Tsub:]*leftweight + aux[:Tsub]*rightweight`	corrected blocks cross-faded with linear weights (Prats 2005, Sec. III-B); the index bookkeeping of the port is not the identity for $\Delta C_0 = 0$ (relative error 0.45, earlier note), the IDL one is
IDL margin	`in_lim = posix + [-Nzp/2, Nzp/2-1]`; `aux = shift(data[in_lim], ...)`	the FFT block is N_{zp} real samples centred on the sub-aperture; only the central $2\,T_{sub}$ are kept
IDL DEM	`heights = ref[posix,3] - dem[posaux*dimr + posvector]`; with `/norcmc`: `posrange = rm*cos(betasub)`	the DEM is read at the azimuth cell of the bin *and* at the range the energy of that bin comes from: before RCMC a scatterer at zero-Doppler range r appears at $r/\cos\beta$

Table 1: The reference kernel.

2.3 The 2-D references

`sata` with `/norcmc` is the reference for data that have not been RCMC'd — our case, since SATA runs on the range-compressed channels before the reconstruction, which does its own RCMC in step 2. `sata1_5d` splits the range spectrum into sub-looks and runs `sata` on each with that sub-look's wavelength, when the relative bandwidth is large enough for $2\pi\Delta C_0/\lambda$ to change across the range band. `sata2d` goes further: the Doppler axis of every range-frequency bin is computed with its own $\lambda(f_r)$, the data are cut into range blocks of 32 samples with margins, and residual range shifts of the motion compensation are applied as ramps. Both are for high squint or very large relative bandwidth; the size of what they add for this system is given in Section 4.5.

3 The implementations

3.1 `sata1d`

Stage A builds the map $\Delta C_0[x]$ of each channel from the scene (Section 4 of the 1D report): the residual $\Delta C_{0,i}(p) = C_{0,i}(p) - C_{0,i}(p_{\text{ref}})$ from the same range-history fit that builds the reconstruction filters; the cell x_t of the scatterer on the channel line including the $b_{at,i}/2$ shift of the phase centre; a block $x_t \pm h_w$ ($h_w = 82$ cells, the second null of the window response, 99.9 % of the energy); the same block at the fold images $x_t + mX$ ($X = 405$ cells) that the under-sampled channel line produces; neighbouring blocks joined and interpolated; overlapping images resolved by the nearest-image rule; zero elsewhere. Stage B is the kernel of Table 1 with the sizing, axis, ruler and rotation of the reference, and the overlap-add of Section 4.1. Per sub-band k, the bins are labelled with the absolute Doppler in $[f_k - \text{PRF}_{\text{ch}}/2, f_k + \text{PRF}_{\text{ch}}/2)$ and the positions are taken on the broadside grid ($\beta_{\text{img}} = 0$).

3.2 `sata2d`

The channel cube $[N_{rx}, N_{a,ch}, N_r]$ is range compressed; SATA corrects every channel line of every active range bin with the same kernel, and the reconstruction (two-step generalized range-Doppler, ported from the reference) follows. What is 2-D is the map: `ResidualMaps` keeps, per channel and scatterer, the bistatic range bin n of closest approach, the azimuth cell (with $b_{at}/2$) and ΔC_0; the map of data range bin m, read by a Doppler bin of label f, holds the scatterers whose energy at that Doppler reaches m (Section 4.4), drawn as the footprint map of Stage A with the fold images of the kernel's centre frequency. Whole band: one pass with $f_{\text{centre}} = 0$; per sub-band: one pass per output band with f_k, the reconstruction keeping the rows of that band.

3.3 Element by element

element	reference	`sata1d` / `sata2d`	status
sub-aperture length	$2\lfloor \delta x \,\text{prf}/2v \rceil$ with the $/N_{sb}$	the same without $/N_{sb}$, prf = rate of the line (PRF_{ch})	equal (bug 1 removed)
FFT size, hop, margin	$N_{\text{zp}} = 2^{\lceil \log_2 T_{\text{sub,eff}} \rceil} \cdot \text{osf}$, hop $T_{\text{sub,eff}}/2$	the same; osf = 4, $N_{\text{zp}} = 128$	equal
Doppler axis	`fsub` rolled, Nyquist bin at +prf/2	`fftfreq`, wrapped into $[f_k - \text{PRF}_{\text{ch}}/2, f_k + \text{PRF}_{\text{ch}}/2)$	equal except the Nyquist bin (1 of 128)
ruler	$r(\tan\beta - \tan\beta_{sq})/v \cdot \text{prf}$	$r \tan\beta / v \cdot \text{PRF}_{\text{ch}}$ (broadside image)	equal for the whole band; bug 3 removed per sub-band
cell lookup	`int()` truncation	rounding	≤ 0.5 cell
rotation	$e^{\mp j(-2\pi\Delta C_0/\lambda)}$	the same	equal (T12, V9)
overlap	rectangular block, cross-fade of corrected blocks	triangular analysis window, constant overlap-add	tested, Section 4.1
margin (IDL)	real samples around the block	zero padding	tested, Section 4.1
map	1D array from outside / DEM per (cell, range)	footprint map of the point scene, with fold images	Section 4.3
range of the lookup	`/norcmc`: DEM read at $r\cos\beta$ per bin	`sata2d`: scatterer bin gated by rcm(f) per Doppler group	added, Section 4.4
$\lambda(f_r)$ (`sata1_5d`, `sata2d`)	per sub-look / per range frequency	centre wavelength	not needed, $\leq 2°$ (Section 4.5)
residual range shifts (`sata2d`)	ramps in the range spectrum	none	not needed: the geometry has no motion error

Table 2: Every element of the reference kernel and what the implementations do with it.

4 Review findings

4.1 Sub-aperture bookkeeping

The reference multiplies the FFT of a rectangular block by the correction and cross-fades the corrected blocks with linear weights; `sata1d` and `sata2d` weight the block with a triangular window *before* the FFT and overlap-add the corrected blocks, normalised by the window sum. Both are "50 % overlap with linear weights"; the difference is whether the window sits before or after the correction, i.e. whether the STFT that assigns bins to cells is that of a rectangular block (sinc, −13 dB sidelobes) or of a triangular one (sinc^2, −26 dB). A third variant, the IDL margin, fills the $N_{\mathrm{zp}} = 128$ samples of the FFT with the real neighbours of the sub-aperture. All three were implemented as the option `scheme` of the kernel (`"cola"`, `"crossfade"`, `"margin"`); all three reproduce the input to 10^{-16} when the map is zero. Table 3 gives the worst ambiguity after the 1D reconstruction on six scenes.

		triangular, COLA		reference cross-fade		IDL margin		
scene	no SATA	whole	sub	whole	sub	whole	sub	oracle
1 target 240 m	−13.1	−39.4	−39.4	−37.2	−37.2	−39.4	−39.4	−39.4
2 targets 3 km apart	−12.8	−38.3	−38.3	−36.8	−36.8	−38.3	−38.3	−38.3
along-track ramp $N = 9$, 20°	−7.0	−30.9	−30.9	−30.7	−30.7	−30.8	−30.8	−30.8
random heights $N = 9$, 500 m	−7.9	−17.2	−17.2	−17.8	−17.8	−15.9	−15.9	−30.8
spike $N = 9$, 500 m	−7.7	−13.1	−13.1	−15.1	−15.1	−16.4	−16.4	−30.8
cliff $N = 9$, 250 m	−12.8	−25.6	−25.6	−24.6	−24.7	−22.3	−22.3	−30.8

Table 3: Worst ambiguity [dB] with the three bookkeeping schemes (SATA 1D, $b_{xt} \sim U(0, 100)$ m, seed 0). Where SATA is inside its validity (isolated targets, the ramp) the triangular window is at the oracle and the rectangular cross-fade is 2 dB behind, because the sinc sidelobes of the rectangular block carry energy into bins whose cells read zero; where neighbours 200 m apart differ by hundreds of metres (spike, random, cliff) all three sit at the resolution limit δx and differ by 1–3 dB either way.

The choice of bookkeeping does not move the result where SATA is valid and does not rescue it where it is not; in the 2-D chain the reference cross-fade and the triangular window agree within 0.2 dB on the scenes of Table 5. The triangular analysis window is kept as the default because it is at the oracle for the isolated target and is the identity by construction.

4.2 Doppler axis and cell rounding

The reference axis and `fftfreq` differ in one bin, the Nyquist one, which the reference labels $+\mathrm{prf}/2$ and the implementation $-\mathrm{prf}/2$; the two labels are the same frequency modulo $\mathrm{PRF}_{\mathrm{ch}}$ and point at fold images X apart, both of which the map carries with the same value for one scatterer. The truncation `int()` of the reference biases the cell by up to one cell towards zero; rounding is used instead. Neither is measurable in the results.

4.3 From a DEM to the footprint map

The reference reads a DEM at the azimuth cell and range a Doppler bin points to; a DEM is defined everywhere, so every bin gets a correction. The implementations work on scenes of point scatterers and build the map from them. The 1D report established what the map must contain for the STFT of a channel line: the value $\Delta C_0(p)$ on the cells where the energy of p appears in the sub-aperture spectra — the block $x_t \pm h_w$ around its own cell, and the same block at $x_t + mX$, the fold images. The folds have no counterpart in the reference: there the data are sampled above the Doppler bandwidth, so a target's energy never wraps, whereas a channel line at $\mathrm{PRF}_{\mathrm{ch}} = 500$ Hz folds the 1281 Hz of the illumination, and a bin labelled f carries energy of true Doppler f and $f \pm \mathrm{PRF}_{\mathrm{ch}}$ (up to $f \pm 2\mathrm{PRF}_{\mathrm{ch}}$ for the outer sub-bands). Where scatterers are closer than $2h_w$ their blocks are joined and interpolated, which is the

point-scene version of reading a DEM; where images of different scatterers overlap the nearest image wins (T17, the collision study). With a real DEM the same code would draw the DEM's residual at every cell and at its fold copies, so the map is a DEM lookup with the folding of the channel line added.

4.4 The range lookup of the 2-D map

Before RCMC the energy of a scatterer of closest-approach range r_n (bin n) at Doppler f lies at $r_n/\cos\beta(f)$, i.e. $\mathrm{rcm}(f) = r_0(1/\cos\beta - 1)/\rho_r$ bins beyond n (up to 1.6 bins at the edge of the beam, $\rho_r = 26$ m). The reference's `/norcmc` branch reads the DEM at $r\cos\beta$ for every bin, so a Doppler bin of data range bin m is corrected with the topography of zero-Doppler range $m - \mathrm{rcm}(f)$. The 25 September version of `sata2d` copied the map of a scatterer to the bins $[n-2, n+4]$ for all Doppler bins and, where two scatterers claimed the same cell of a bin, took the one whose energy centroid was nearest (kept as `rcm_lookup=False`). That is right for isolated scatterers and approximate when scatterers at the same azimuth sit one or two bins apart in range — the cross-track ramps at $\alpha = 10$–$15°$ with $N \geq 7$. The review replaced it with the reference rule: the Doppler bins of the kernel are grouped into 16 groups of $\mathrm{PRF_{ch}}/16 = 31$ Hz; for group g with representative label f_g, each image m_{img} of the footprint map is drawn from the scatterers with $|n + \mathrm{rcm}(f_g - m_{\mathrm{img}}\mathrm{PRF_{ch}}) - m| \leq$ gate (the true Doppler behind the fold image at $+m_{\mathrm{img}}X$ is $f_g - m_{\mathrm{img}}\mathrm{PRF_{ch}}$), nearest in range first. The kernel takes the stack of 16 maps and a row index per bin (Fig. 1). The gate covers the compressed pulse and its skirts; Table 4 chooses it.

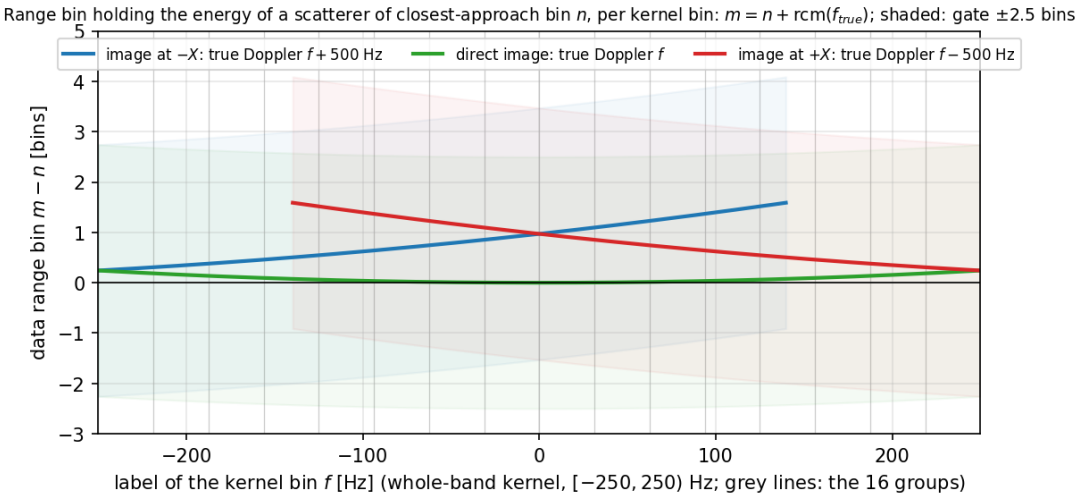


Figure 1: The range side of the lookup. A kernel bin labelled f holds, for a scatterer of closest-approach bin n, energy of true Doppler f (direct image at x_t) and $f \pm \mathrm{PRF_{ch}}$ (fold images at $x_t \mp X$), each lying $\mathrm{rcm}(f_{true})$ bins beyond n. Data range bin m therefore reads, in that bin, only the images whose curve passes within the gate of m; the bins are grouped in 16 (grey) so that 16 maps per data range bin suffice.

gate [bins]	1.5	2.0	2.5	3.0	4.0
1 target 240 m	−36.6	−37.8	−39.5	**−39.5**	−39.5
1 target 400 m	−33.2	−34.4	−39.1	**−39.6**	−39.5
cross-track $N = 9$, 10°	−32.3	−32.4	−32.5	**−32.5**	−32.4
cross-track $N = 9$, 15°	−28.9	−29.0	−29.0	**−29.0**	−28.9
cross-track $N = 13$, 15°	−31.1	−31.2	−31.2	**−31.2**	−31.2
cross-track $N = 5$, 30°	−32.2	−33.3	−34.7	**−34.7**	−34.8

Table 4: Worst ambiguity [dB] of the whole-band SATA 2D against the range gate of the lookup (band-limited generator). Below 2.5 bins the compressed pulse's own skirts are left uncorrected; 3 bins is used.

scene	reference	no SATA	map per bin	RCM lookup	oracle
1 target, 240 m	−39.2	−20.1	−39.5	−39.5	−39.5
1 target, 400 m	−39.2	−16.0	−39.5	−39.6	−39.5
cross-track ramp $N = 9$, 10°	−39.2	−17.1	−30.4	−32.5	−36.6
cross-track ramp $N = 9$, 15°	−39.1	−11.8	−25.4	−29.0	−36.3
cross-track ramp $N = 13$, 15°	−39.2	−14.0	−28.3	−31.2	−37.2
cross-track ramp $N = 13$, 5°	−39.4	−21.3	−29.7	−30.2	−31.9
cross-track ramp $N = 5$, 30°	−39.3	−13.7	−35.9	−34.7	−39.3
along-track ramp $N = 9$, 20°	−34.4	−13.9	−31.4	−31.4	−31.3

Table 5: Whole-band SATA 2D with one map per data range bin (the 25 September rule) and with the Doppler-dependent lookup of the reference (gate 3 bins), worst ambiguity [dB], band-limited generator. The lookup gains 2–4 dB where scatterers at the same azimuth are one or two range bins apart and is neutral elsewhere; a gap to the oracle remains in those cases because inside one range bin two such scatterers still share Doppler bins.

4.5 Not implemented, and by how much

Wavelength across the range band (`sata1_5d`, `sata2d`). The correction is $2\pi\Delta C_0/\lambda$; over the sampled range band (±2.9 MHz on 1199 MHz, ±0.24 %) it changes by ±0.24 % of itself. The largest residual of the sweep is $\Delta C_0 = 0.54$ m ($b_{xt} = 64$ m, $h = 2383$ m: the 50° cross-track ramp), a rotation of 778°, so the error at the band edge is 1.9°; for the reference target (240 m, −5.5 cm, 79°) it is 0.2°. The residuals left in the sweep are 20–40 dB below the signal, i.e. phases of degrees, so this term is below what is measured and is not implemented. *The Doppler centroid across the range band* (the reason for `sata2d`) is zero at broadside. *Residual range shifts* are motion-compensation terms of an airborne track; the simulated orbits are straight. *The margin* of the IDL block was tested (Table 3): no gain. *Per-sub-band residual fit*: fitting ΔC_0 over the look angles of each band instead of the whole aperture changes it by 10^{-6} mm (1D report, Section 5.3), so `sata2d` uses one fit.

4.6 The floor of the 2-D chain: found in the signal generator

The first run of the 2-D sweep (25 September) showed a floor that did not belong to SATA: with no topography at all ($h = 0$, map zero) the chain gave a worst ambiguity of −36 dB for a target at the reference range, −33 dB for one 684 m away in slant range, −26 to −29 dB for 13 flat targets spread over 2 km of ground range, and the phase of the reconstructed spectrum against the reference oscillated by ±50° along range frequency even for the oracle. The oracle shares the floor, so it is upstream of SATA; the two-step reconstruction was the suspect. The review traced it further upstream, to the signal generator ported from the reference (`echo_block`): it samples the linear-FM pulse in fast time, and with the preset's pulse duration of 1 μs and range sampling of 5.75 MHz a pulse is 5.7 samples long with hard edges. A receiver displaced

by b_{xt} sees the target $b_{xt} \sin\theta/2$ closer (10.9 m = 0.42 samples for $b_{xt} = 63.7$ m), so its pulse covers a different set of fast-time samples than the reference's: the sampled channel is not the delayed copy of the sampled reference that the reconstruction models, and no filter can make it one. Table 6 measures it directly, channel by channel at the full PRF, against the model the reconstruction assumes, $S_i = S_{\text{mono}}\, e^{-2\pi j\,(C_{0,i}/\lambda(f_r)+(D_{t,i}-C_{1,i})f_a+C_{2,i}f_a^2\lambda)}$.

pulse of the generator	ch 0 (b_{xt} 63.7 m, b_{at} 0) error / phase rms, max	ch 3 (b_{xt} 1.7 m, b_{at} 23.1 m) error / phase rms, max	flat, r_{ref} no SATA (ref.)	flat, +684 m no SATA (ref.)
chirp, 1 μs (5.7 samples)	−6.4 dB / 27.5°, 74°	−17.5 dB / 31.5°, 147°	−37.8 (−35.9)	−32.8 (−39.0)
chirp, 20 μs (115 samples)	−19.0 dB / 16.0°, 97°	−30.3 dB / 0.8°, 3.3°	−39.5 (−39.3)	−39.4 (−39.3)
band-limited, exact delay	−65.1 dB / 0.03°, 0.05°	−31.3 dB / 0.14°, 0.21°	−39.5 (−39.2)	−39.4 (−39.2)

Table 6: The signal generator against the model of the reconstruction (one flat target, $b_{xt} \sim U(0, 100)$ m). Left: error-to-signal of the full-PRF channel relative to "monostatic × model", in band, and the phase of channel over model along f_r at $f_a = 0$. Right: worst ambiguity of the reconstruction without SATA for a flat target at and away from the reference range. The along-track channel (ch 3) keeps a −31 dB residual with any pulse: the antenna window of the receive leg is shifted by b_{at}/v, an amplitude effect at the aperture edges that the model does not carry (phase along f_a 1.1° rms).

A longer pulse dilutes the edge effect (time-bandwidth 100 instead of 5); the fix adopted removes it: the generator now builds each echo in the range-frequency domain, as the band-limited pulse of bandwidth r_{bw} with the chirp phase that `range_compress` removes, delayed by the exact $2R/c$ through the shift theorem (`Params3D.pulse = "ideal"`; the sampled chirp is kept as `"chirp"`). Sampled channels are then exact delayed copies of one another, the reconstruction of a flat target is at the reference level at every range, and the phase panels of Section 9 show the reconstruction's own error instead of the generator's. The 1-D chain has no range axis and is untouched. All 2-D numbers published before this document — the case matrix, the co-registration study, the 25 September sweep — carry the artifact; the sweep of Section 7 is re-run with the corrected generator, and the range lookup of Section 4.4 was re-tuned on it (gate 3 bins).

5 Verification

5.1 SATA 1D — 19 checks

	what is checked	result
T1–T3	zero map = identity; constant map = one global rotation; inject/remove round trip	10^{-16}, 10^{-16}, 0.8 %
T4, T5	residual vs closed form $-b_{xt}\Delta h/(r_0 \tan\theta)$; scatterer cell vs simulated closest approach	0.998–1.000; 0 cells
T6, T7	STFT peak $\to x_t + mX$ (whole band); own-band peak $\to x_t$ (per sub-band)	2.4 cells (<1 bin)
T8, T9	energy of the target read inside the map; corrected channel vs ideal rotation	99.9 %; 2.0 %
T10	one target, $h = 0 \ldots 400$ m: ambiguity vs reference	-39.4 vs -39.1 dB (no SATA $-23 \ldots -9$)
T12	sign: remove / inject / remove twice	-39.4 / -7.8 / -13.1 dB
T13	two-target map applied to each target alone reads its own ΔC_0	2.2 %
T11, T14, T17a	multi-target scenes vs the oracle	within 0.3 dB when resolvable
T15, T16	footprint null (1st–4th), FFT oversampling 1–8	same ambiguity
T17b	two targets vs separation: Doppler-alias collision at $\lvert d - mX\rvert \lesssim 0.6$ km	inherent; outside: oracle
T18, T19	$b_{xt} = 0 \Rightarrow$ identity; Doppler axes (kernel and sub-bands)	5×10^{-11}; exact

Table 7: `sata1d_code/verify_sata1d.py`, all pass.

5.2 SATA 2D — 15 checks

	what is checked	result
V1	kernel axis per band: whole = `fftfreq`; band k within $[f_k - \mathrm{PRF_{ch}}/2, f_k + \mathrm{PRF_{ch}}/2)$, aliases of the baseband bins	exact
V2	zero map = identity; constant map = rotation	1.1×10^{-16}, 1.4×10^{-16}
V3	scatterer cell on the channel line, closest approach of the bistatic pair ($b_{at}/2$)	≤ 0.5 cell
V4	range bin and migration: 96–97 % of the channel energy in $[n-2, n+4]$, centroid $+0.3 \ldots +1.2$ bin beyond n	pass
V5	map row: blocks $x_t \pm 82$ at $m = \{-1, 0, 1\}$ (whole), $\{-1..2\}$ (band 2), active bins $[n-2, n+4]$	exact
V6	energy of the target read inside the map, per channel	99.94–99.95 %
V7	corrected channel vs ideal rotation $e^{+j2\pi\Delta C_0/\lambda}$	1.9, 0.6, 0.1, 0.03 %
V8	one target, $h = 0, 100, 240, 400$ m: no SATA -39.5, -27.9, -20.1, -16.0; SATA at the level of the flat case	-39.5 dB at every height (reference -39.2)
V9	sign: remove / inject / remove twice	-39.5 / -14.7 / -20.1 dB
V10	two targets 3 km apart read their own ΔC_0	0.04, 0.03 %
V11	cross-track ramp (5°): five range bins, each with its own ΔC_0	bins 27, 32, 36, 41, 45
V12	$b_{xt} = 0$: map 10^{-9} mm, channel unchanged	1.2×10^{-11}
V13	whole band vs per sub-band on three iso-range scenes; both vs the oracle	≤ 0.1 dB; within 0.3 dB of the oracle
V14	range lookup: bins gated per Doppler group as $\lvert n + \mathrm{rcm}(f) - m\rvert \leq 3$; one target with / without the lookup	exact; -39.5 / -39.5 dB
V15	signal model: every channel at the full PRF equals the monostatic reference times $e^{-2\pi j(C_0/\lambda(f_r)+(D_t-C_1)f_a+C_2 f_a^2\lambda)}$ (Section 4.6)	error -65, -35, -33, -31 dB

Table 8: `sata2d/verify_sata2d.py`, all pass (log in `sata2d_verification/verify_sata2d_log.txt`).

6 The sweeps

Both algorithms run the same 259 cases: $N_{rx} = 4$, PRF = 2000 Hz, along-track baselines on the DPCA condition ($\Delta b_{at} = 7.69$ m), $b_{xt} \sim U(0, 100)$ m with seed 0 ($b_{xt} = 63.7, 27.0, 4.1, 1.7$ m), $r_0 = 766.2$ km, $\theta = 20°$, $\lambda = 0.25$ m. Nine families of iso-range targets 200 m apart in azimuth with a height profile (Table 9), and the cross-track ramps. Every case is reconstructed without SATA, with SATA whole band, with SATA per sub-band and with the oracle; the reference is the monostatic signal at the full PRF.

family	id	geometry	cases
along-track ramp	RA	$N \in \{2, 3, 5, 7, 9, 13\}$ targets, $h_j = j \cdot 200 \tan\alpha$, $\alpha \in \{0.1, 1, 1.5, 2, 3, 4, 5, 10, 15, 20, 30, 40, 50\}°$	78
zig-zag	ZI	$N \in \{5, 9\}$, slopes $+\alpha, -\alpha, +\alpha, \ldots$	26
piecewise	PW	$N \in \{5, 9, 13\}$, random slopes drawn from the α list with random sign, seeds 0–4	15
steepening / flattening	ST / FL	$N \in \{5, 9\}$, slopes $0.1° \to \alpha_{max}$ (and reversed), $\alpha_{max} \in \{10, 20, 30, 50\}°$	8 + 8
cliff	CL	$N \in \{5, 9\}$, second half raised by $\Delta h \in \{10, 50, 100, 250, 500, 1000, 2000\}$ m	14
spike	SK	$N \in \{5, 9\}$, middle target raised by $\Delta h \in \{100, 500, 1000, 2000\}$ m	8
mesa	ME	$N \in \{5, 9\}$, all but the two ends raised by $\Delta h \in \{100, 500, 1000\}$ m	6
random heights	RH	$N \in \{5, 9, 13\}$, $h \sim U(0, h_{max})$, $h_{max} \in \{100, 500, 1000\}$ m, seeds 0, 1	18
cross-track ramp	TR	$N \in \{2, 3, 5, 7, 9, 13\}$ targets at azimuth 0, $\Delta y = 200 \ldots 2000$ m, $\Delta h = \Delta y \tan\alpha$, same α list	78

Table 9: The 259 cases. Iso-range: each target is displaced cross-track onto $y(h) = \sqrt{r_0^2 - (H - h)^2}$ so that all targets of a case share the slant range r_0 and one range bin; the cross-track ramp is the only family whose targets sit in different range bins (except at $\alpha = \theta = 20°$, where $\Delta h = \Delta y \tan\theta$ keeps them iso-range).

Metric. Worst ambiguity: the largest value of the focused response outside ±40 samples (±154 m) of every target, in dB relative to the peak of the reference. *SATA 1D*: the reconstructed line is correlated with the reference chirp of the scene centre. *SATA 2D*: every target is focused with its own 2-D matched filter (the monostatic signal of that target, so the range cell migration is followed exactly), the azimuth cut through its range row is placed at its position, and the envelope over targets is taken; the reference is treated identically. The 2-D metric is milder on the uncorrected result — an ambiguity is a copy of the target shifted in Doppler, with a different migration, which the 2-D filter of the true target smears in range — so "no SATA" reads about 5–10 dB better in 2D than in 1D for the same scene, and the gains "no SATA − SATA" are correspondingly smaller. The gap between SATA and the oracle is the comparable number.

Improvement heatmaps. For every family, improvement = worst ambiguity (no SATA) − worst ambiguity (SATA), N targets against the family parameter; positive is better. Whole band and per sub-band are identical within 0.1 dB in every one of the 518 runs (the residual is angle-independent for iso-range targets), so the heatmaps show the whole band and the per-sub-band column of the tables says the same.

7 Results by family

Each of the following pages shows one family: its height profiles for $N = 9$ (one curve per heatmap column), the same cases in 3D — azimuth, ground-range offset, height, with the iso-range placement visible as the higher targets sitting further out in ground range — and the improvement heatmaps of SATA 1D and of SATA 2D over the whole $N\times$ parameter grid.

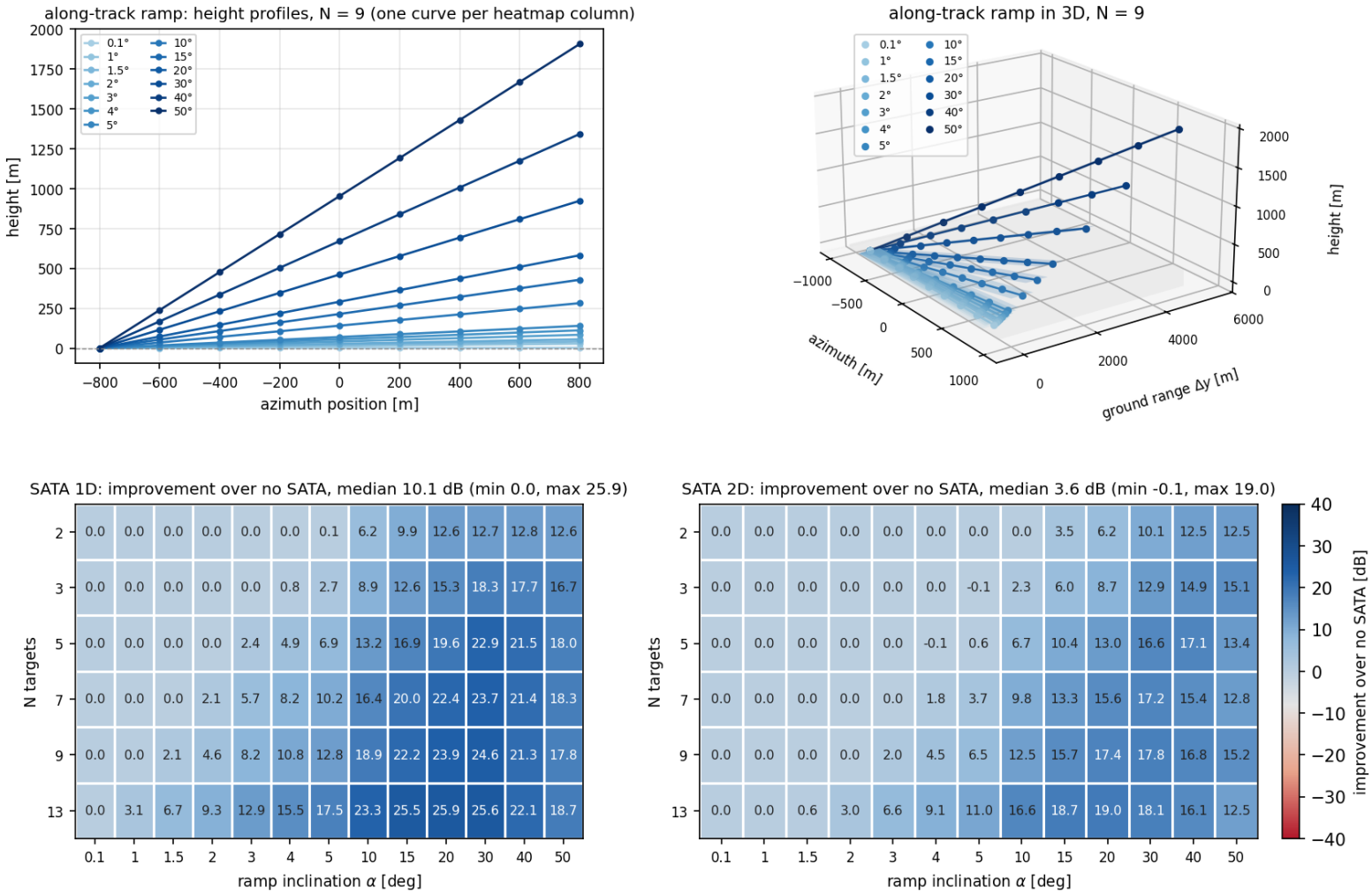


Figure 2: Along-track ramp (78 cases). Top: the height profiles of the $N = 9$ cases and the same targets in 3D (azimuth, ground-range offset, height; iso-range targets sit further out in ground range the higher they are). Bottom: improvement of SATA whole band over no SATA, SATA 1D (median 10.1 dB, gap to the oracle median 0.0, max 10.8 dB) and SATA 2D (median 3.6 dB, gap to the oracle median 0.0, max 6.5 dB).

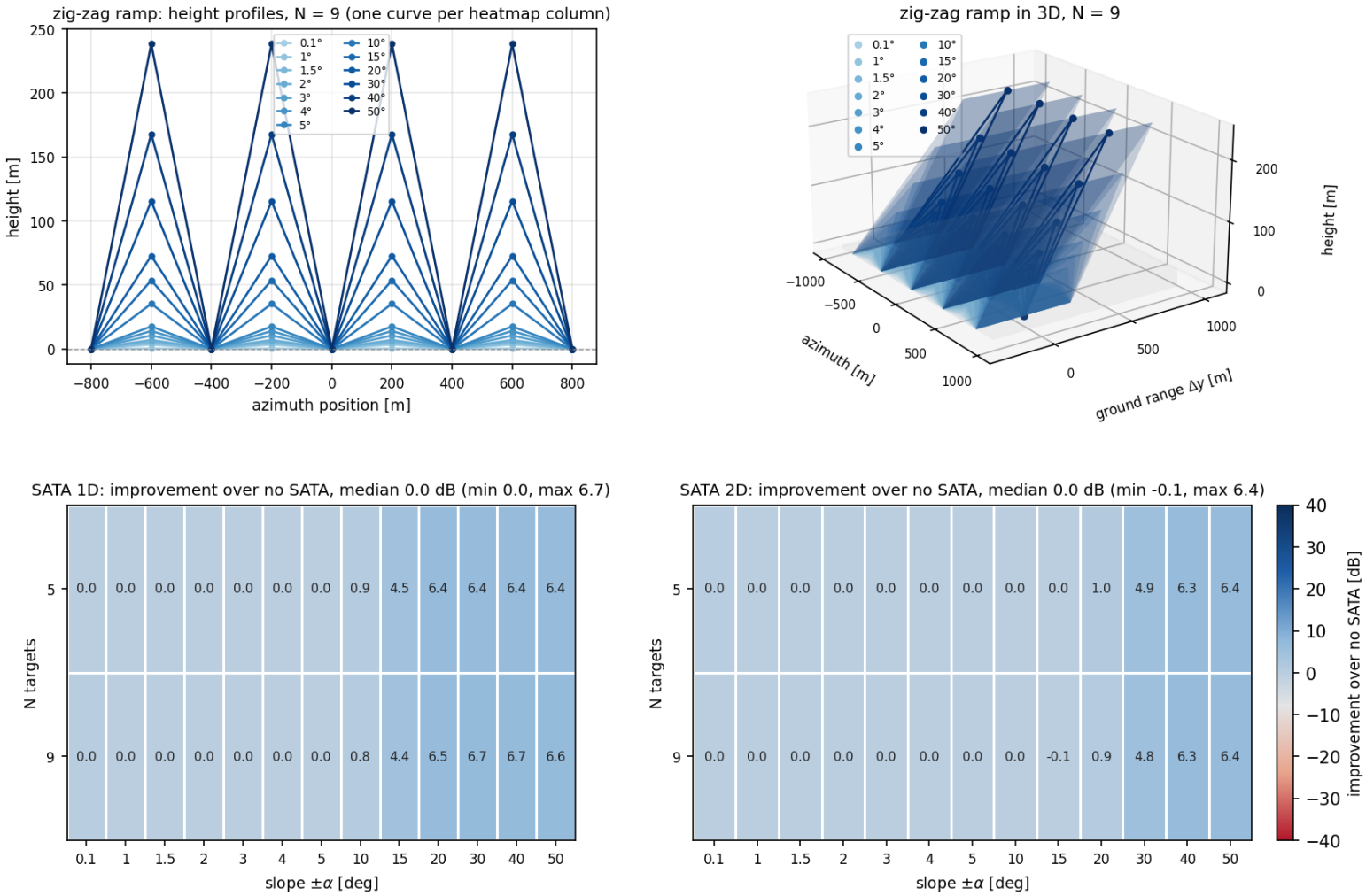


Figure 3: Zig-zag ramp (26 cases). Top: the height profiles of the $N = 9$ cases and the same targets in 3D (azimuth, ground-range offset, height; iso-range targets sit further out in ground range the higher they are). Bottom: improvement of SATA whole band over no SATA, SATA 1D (median -0.0 dB, gap to the oracle median 0.0, max 11.5 dB) and SATA 2D (median -0.0 dB, gap to the oracle median 0.0, max 5.0 dB).

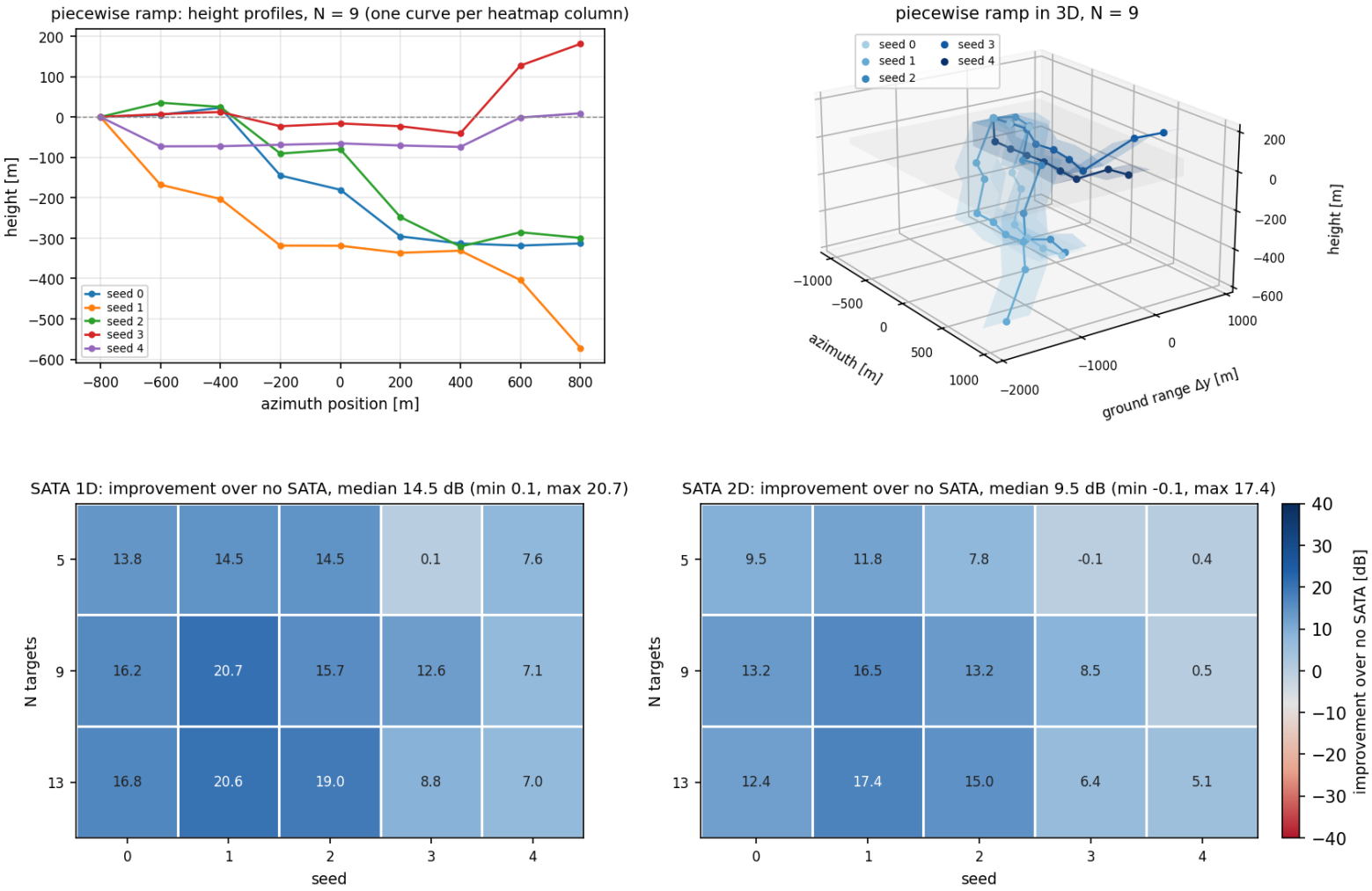


Figure 4: Piecewise ramp (15 cases). Top: the height profiles of the $N = 9$ cases and the same targets in 3D (azimuth, ground-range offset, height; iso-range targets sit further out in ground range the higher they are). Bottom: improvement of SATA whole band over no SATA, SATA 1D (median 14.5 dB, gap to the oracle median 3.1, max 7.4 dB) and SATA 2D (median 9.5 dB, gap to the oracle median 0.1, max 2.6 dB).

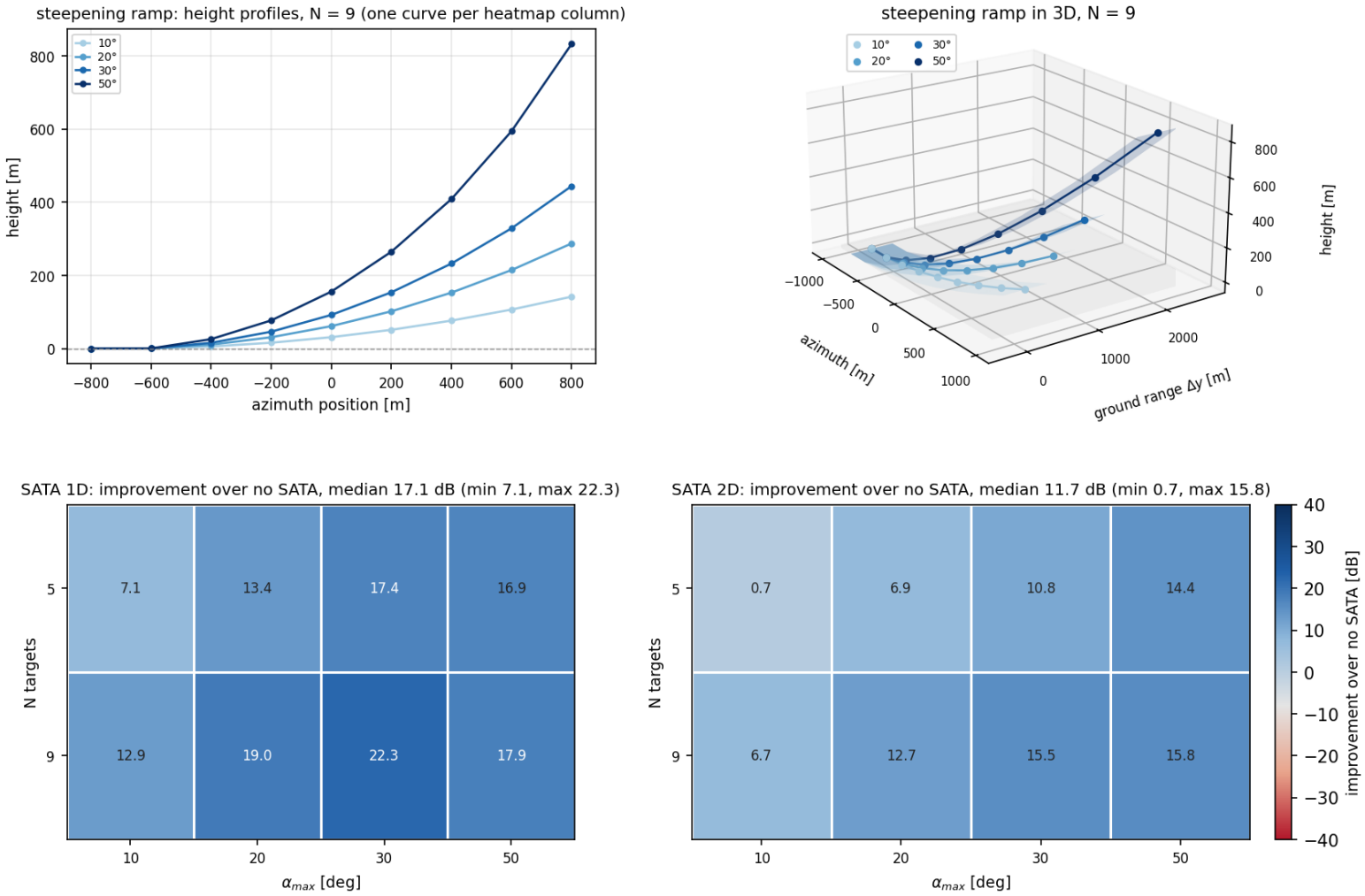


Figure 5: Steepening ramp (8 cases). Top: the height profiles of the $N = 9$ cases and the same targets in 3D (azimuth, ground-range offset, height; iso-range targets sit further out in ground range the higher they are). Bottom: improvement of SATA whole band over no SATA, SATA 1D (median 17.1 dB, gap to the oracle median -0.0, max 6.0 dB) and SATA 2D (median 11.7 dB, gap to the oracle median 0.1, max 1.9 dB).

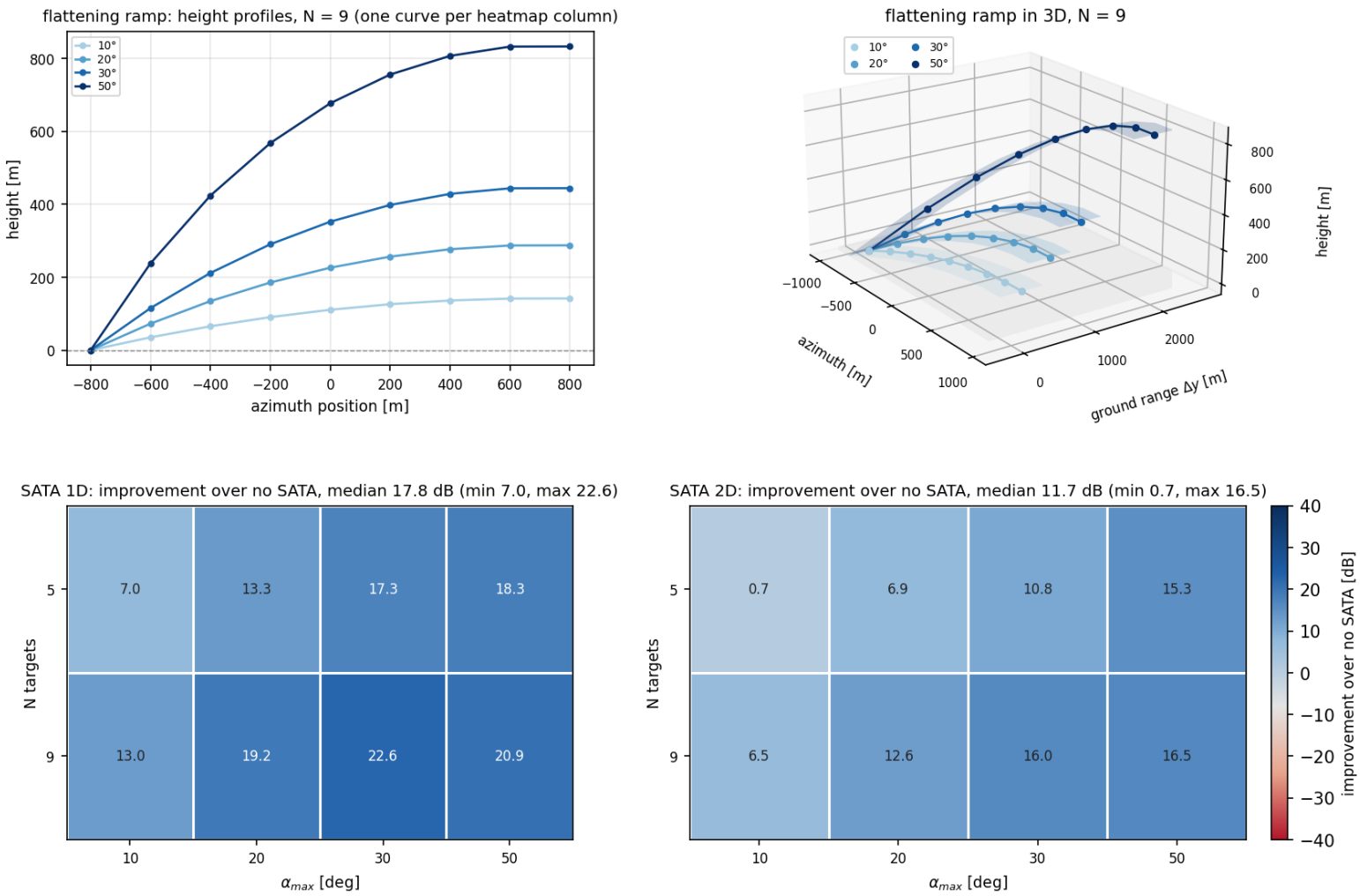


Figure 6: Flattening ramp (8 cases). Top: the height profiles of the $N = 9$ cases and the same targets in 3D (azimuth, ground-range offset, height; iso-range targets sit further out in ground range the higher they are). Bottom: improvement of SATA whole band over no SATA, SATA 1D (median 17.8 dB, gap to the oracle median 0.1, max 4.2 dB) and SATA 2D (median 11.7 dB, gap to the oracle median 0.1, max 1.3 dB).

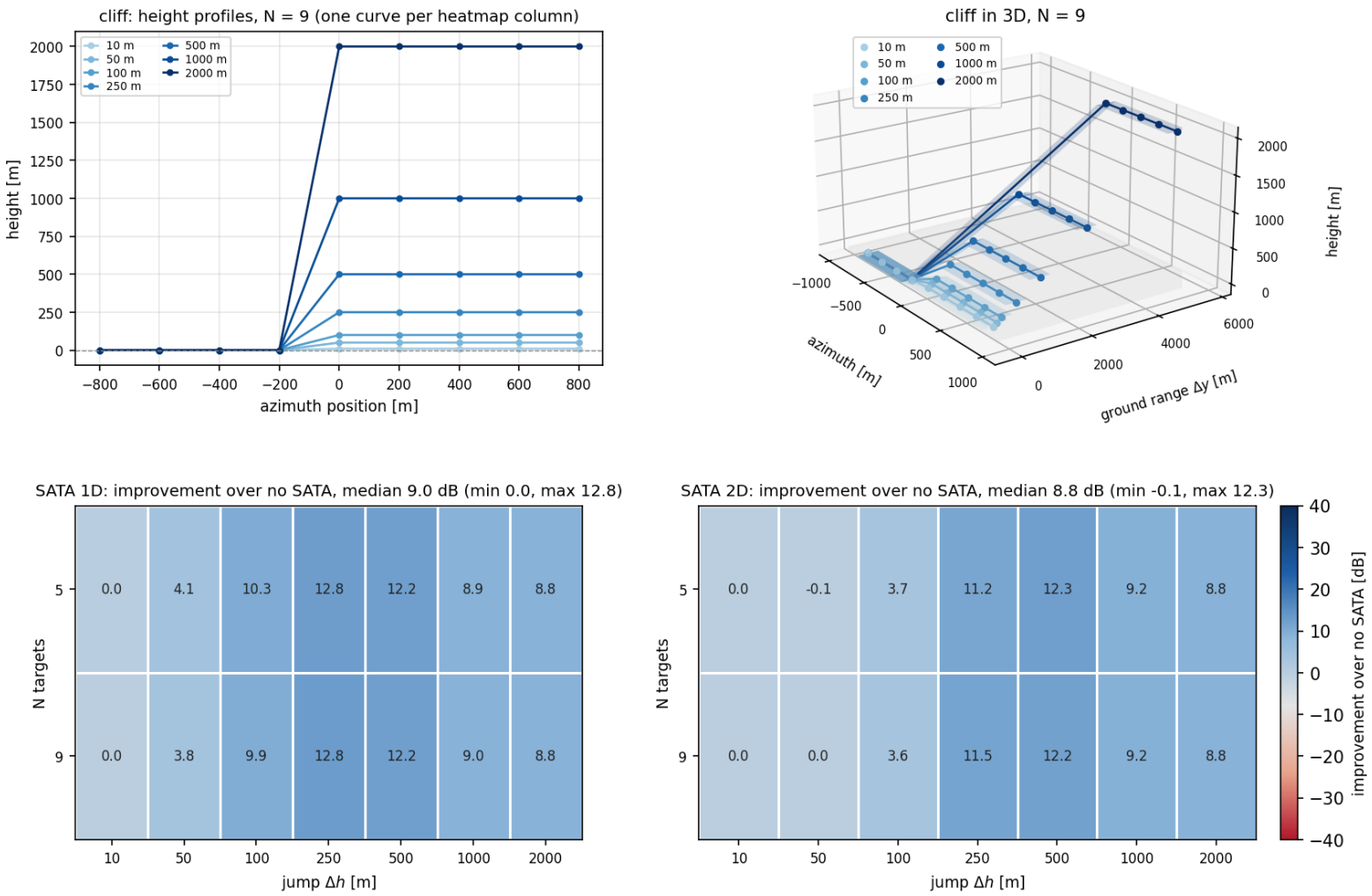


Figure 7: Cliff (14 cases). Top: the height profiles of the $N = 9$ cases and the same targets in 3D (azimuth, ground-range offset, height; iso-range targets sit further out in ground range the higher they are). Bottom: improvement of SATA whole band over no SATA, SATA 1D (median 9.0 dB, gap to the oracle median 5.3, max 14.0 dB) and SATA 2D (median 8.8 dB, gap to the oracle median 0.3, max 7.3 dB).

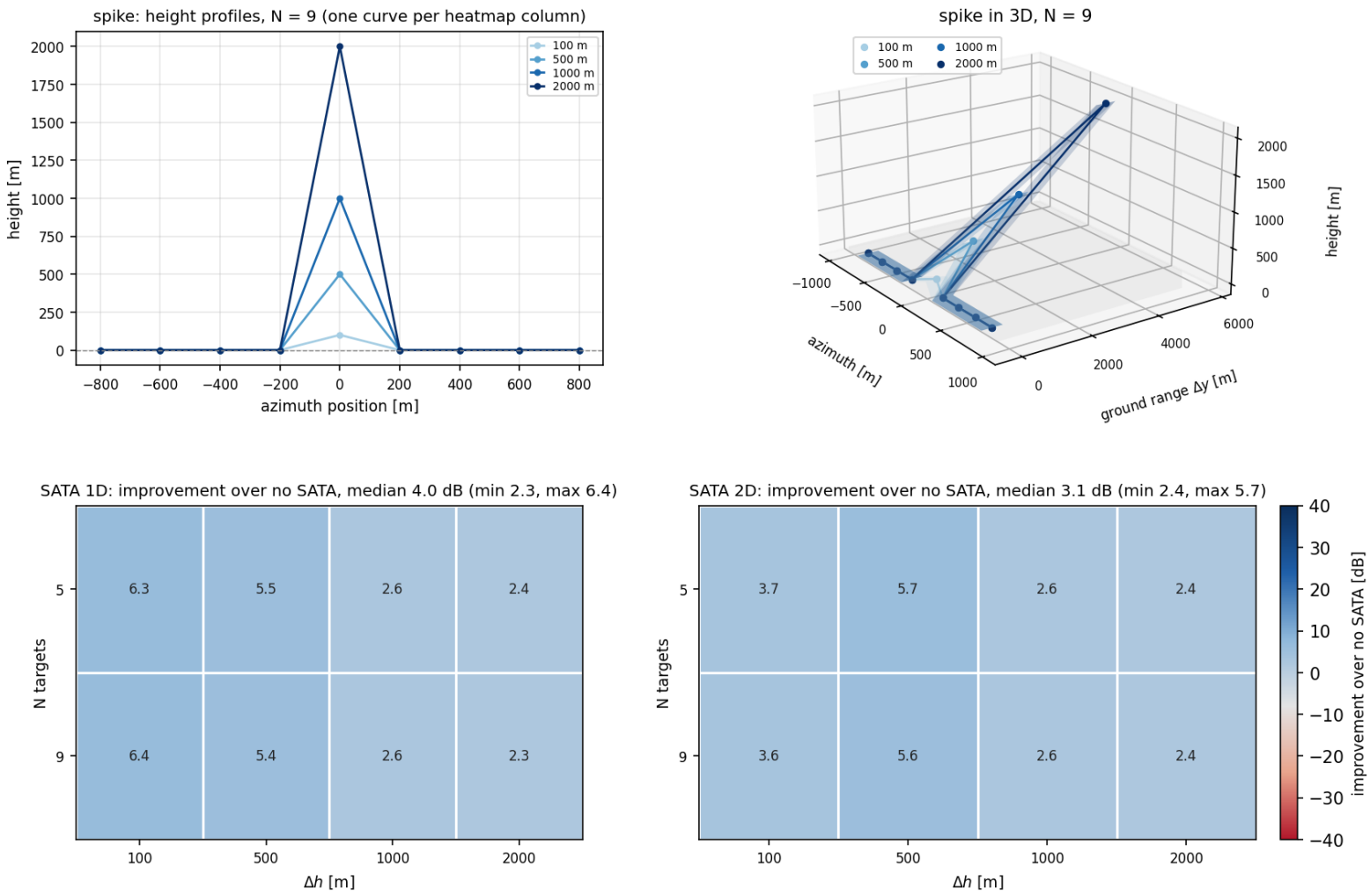


Figure 8: Spike (8 cases). Top: the height profiles of the $N = 9$ cases and the same targets in 3D (azimuth, ground-range offset, height; iso-range targets sit further out in ground range the higher they are). Bottom: improvement of SATA whole band over no SATA, SATA 1D (median 4.0 dB, gap to the oracle median 18.9, max 20.3 dB) and SATA 2D (median 3.1 dB, gap to the oracle median 12.3, max 13.7 dB).

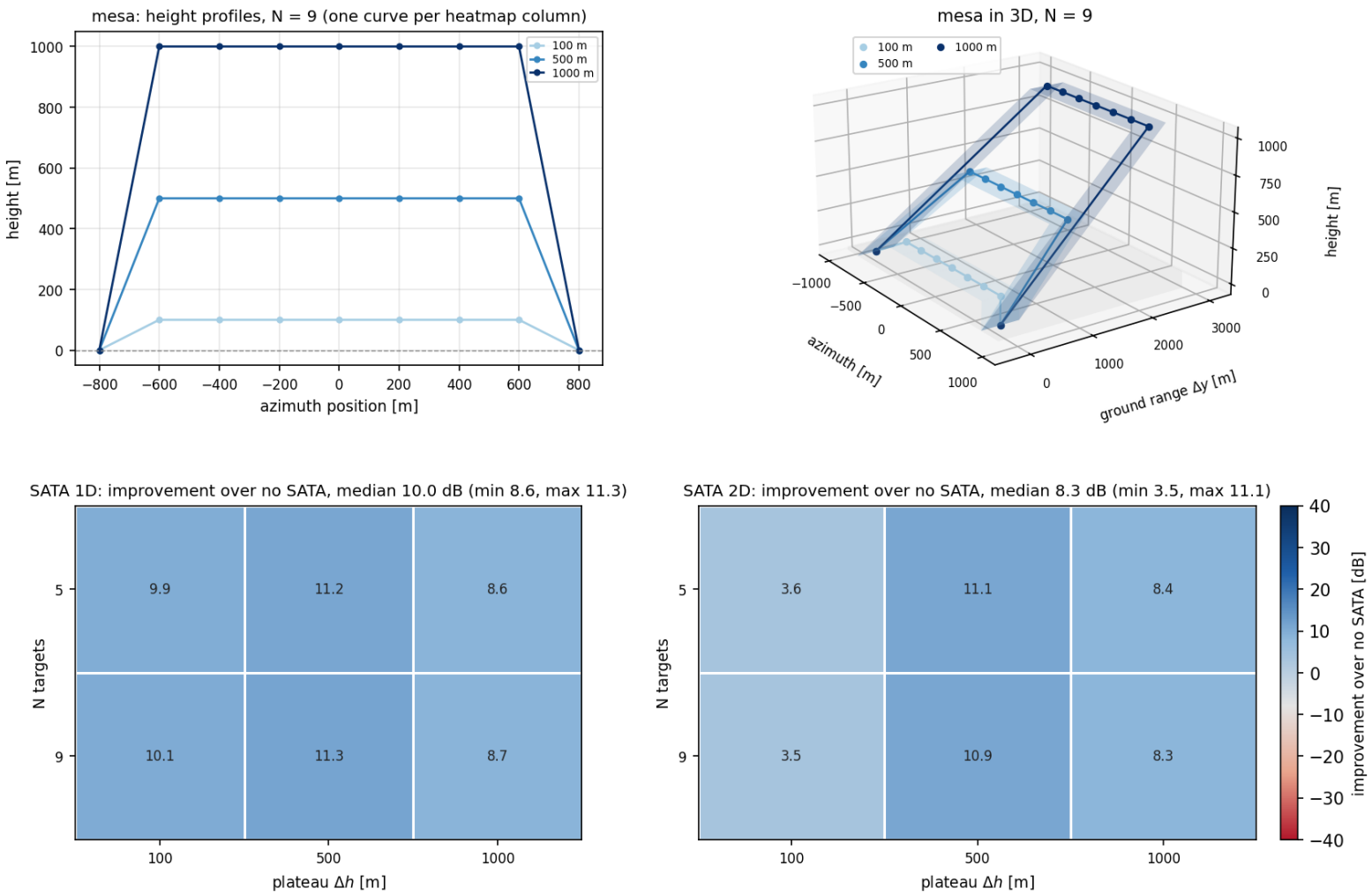


Figure 9: Mesa (6 cases). Top: the height profiles of the $N = 9$ cases and the same targets in 3D (azimuth, ground-range offset, height; iso-range targets sit further out in ground range the higher they are). Bottom: improvement of SATA whole band over no SATA, SATA 1D (median 10.0 dB, gap to the oracle median 12.2, max 14.3 dB) and SATA 2D (median 8.3 dB, gap to the oracle median 5.7, max 7.9 dB).

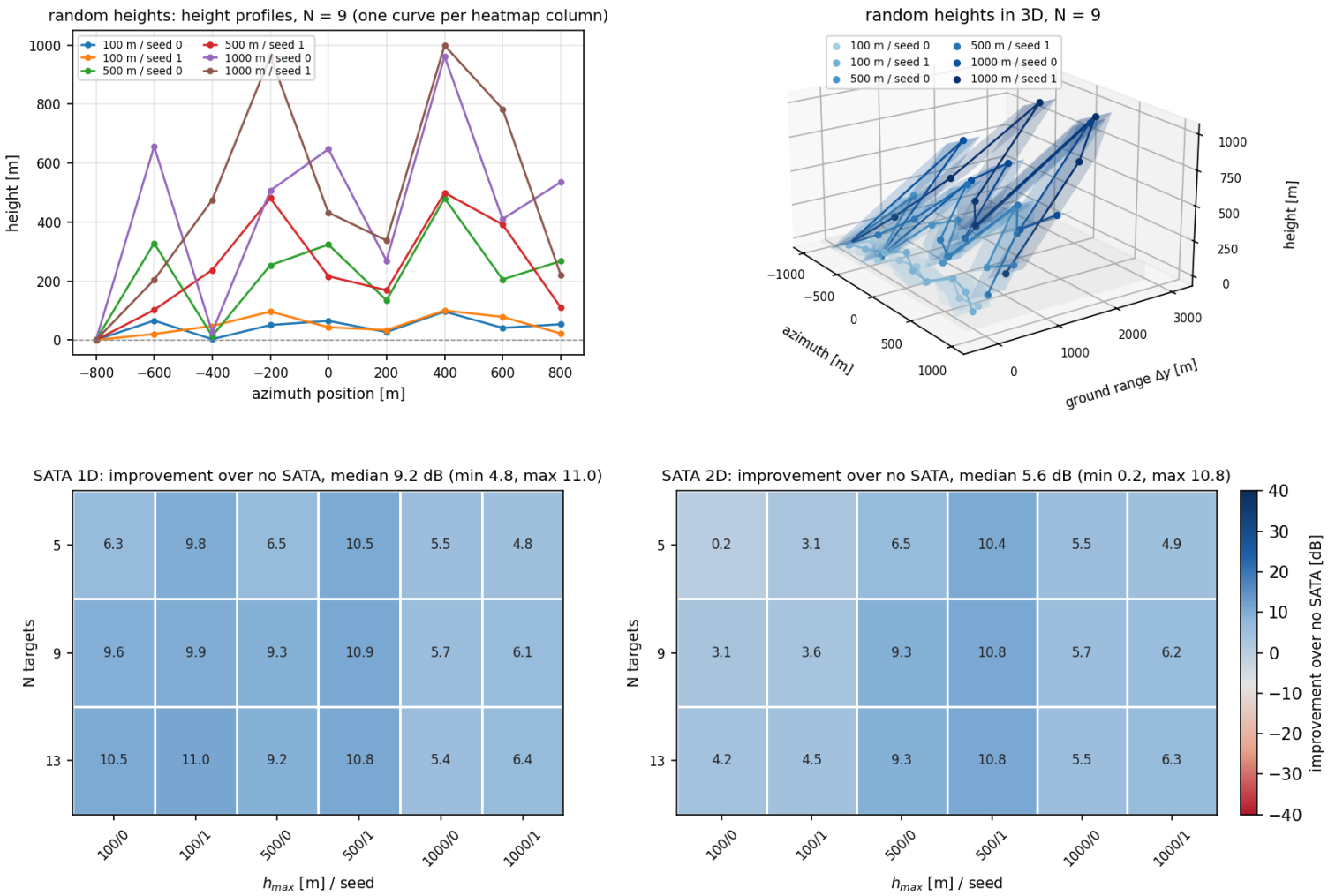


Figure 10: Random heights (18 cases). Top: the height profiles of the $N = 9$ cases and the same targets in 3D (azimuth, ground-range offset, height; iso-range targets sit further out in ground range the higher they are). Bottom: improvement of SATA whole band over no SATA, SATA 1D (median 9.2 dB, gap to the oracle median 13.5, max 20.0 dB) and SATA 2D (median 5.6 dB, gap to the oracle median 7.0, max 13.3 dB).

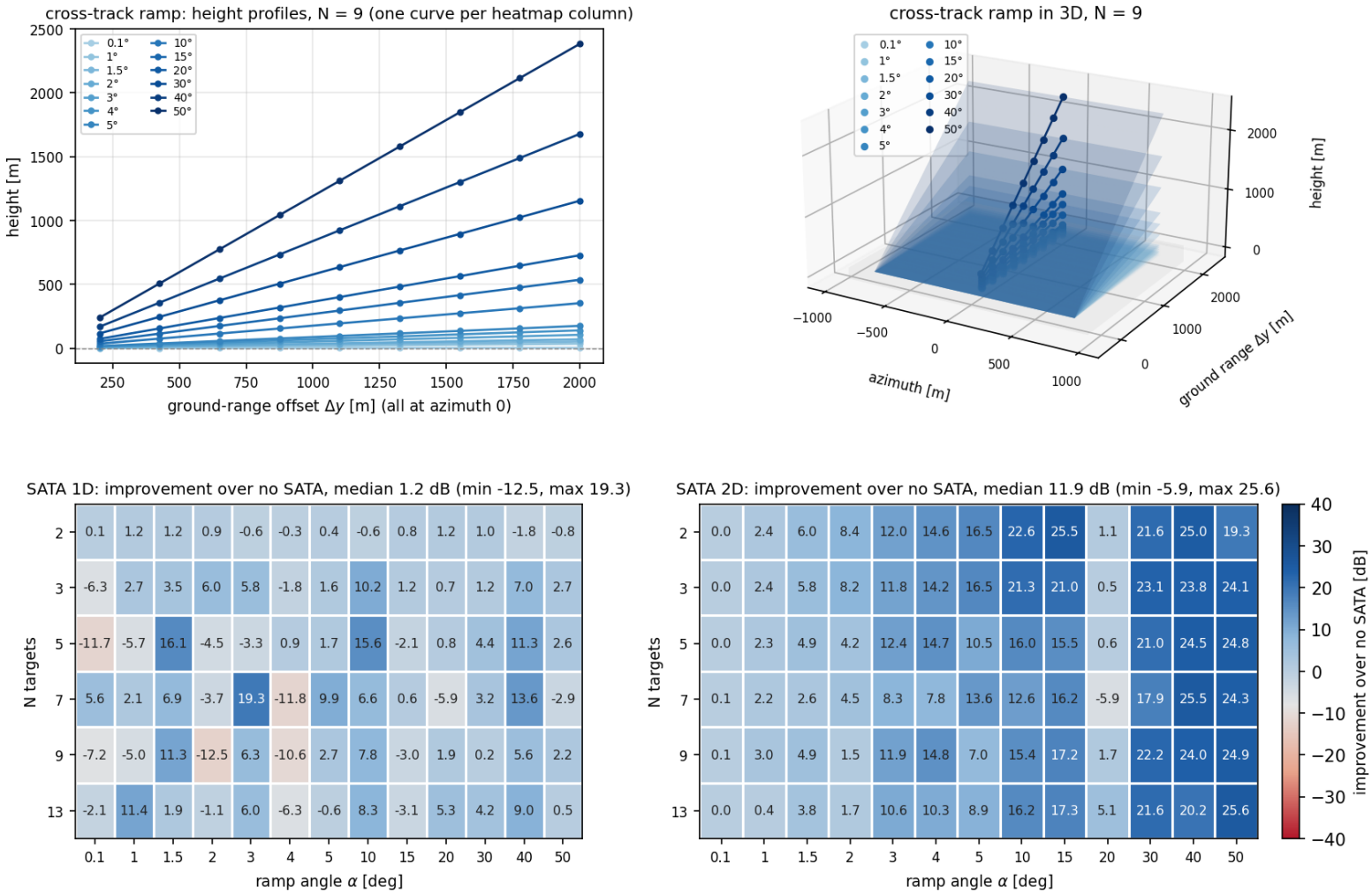


Figure 11: Cross-track ramp (78 cases). Top: the height profiles of the $N = 9$ cases and the same targets in 3D (azimuth, ground-range offset, height; iso-range targets sit further out in ground range the higher they are). Bottom: improvement of SATA whole band over no SATA, SATA 1D (median 1.2 dB, gap to the oracle median 29.2, max 46.0 dB) and SATA 2D (median 11.9 dB, gap to the oracle median 0.3, max 25.5 dB).

8 Summary tables

family	alg.	cases	reference	no SATA	SATA whole	SATA sub	oracle	gain	gap to oracle
Along-track ramp	1D	78	−34.5	−22.5	−31.0	−31.0	−31.4	10.1 (0.0..25.9)	0.0 (max 10.8)
	2D	78	−34.7	−29.2	−31.5	−31.5	−32.0	3.6 (-0.1..19.0)	0.0 (max 6.5)
Zig-zag ramp	1D	26	−34.1	−30.8	−30.8	−30.8	−30.9	0.0 (0.0..6.7)	0.0 (max 11.5)
	2D	26	−34.4	−31.3	−31.3	−31.3	−31.4	0.0 (-0.1..6.4)	0.0 (max 5.0)
Piecewise ramp	1D	15	−33.9	−14.1	−28.3	−28.2	−31.0	14.5 (0.1..20.7)	3.1 (max 7.4)
	2D	15	−34.4	−21.3	−31.4	−31.4	−31.5	9.5 (-0.1..17.4)	0.1 (max 2.6)
Steepening ramp	1D	8	−34.1	−12.8	−30.8	−30.8	−30.9	17.1 (7.1..22.3)	0.0 (max 6.0)
	2D	8	−34.4	−19.6	−31.3	−31.3	−31.4	11.7 (0.7..15.8)	0.1 (max 1.9)
Flattening ramp	1D	8	−34.1	−12.7	−30.8	−30.8	−30.9	17.8 (7.0..22.6)	0.1 (max 4.2)
	2D	8	−34.4	−19.7	−31.3	−31.3	−31.4	11.7 (0.7..16.5)	0.1 (max 1.3)
Cliff	1D	14	−34.1	−12.7	−25.6	−25.6	−30.9	9.0 (0.0..12.8)	5.3 (max 14.0)
	2D	14	−34.4	−19.8	−31.1	−31.1	−31.4	8.8 (-0.1..12.3)	0.3 (max 7.3)
Spike	1D	8	−34.1	−8.3	−11.9	−11.9	−30.9	4.0 (2.3..6.4)	18.9 (max 20.3)
	2D	8	−34.4	−15.3	−19.1	−19.1	−31.4	3.1 (2.4..5.7)	12.3 (max 13.7)
Mesa	1D	6	−34.1	−8.0	−18.7	−18.7	−30.9	10.0 (8.6..11.3)	12.2 (max 14.3)
	2D	6	−34.4	−15.3	−25.7	−25.7	−31.4	8.3 (3.5..11.1)	5.7 (max 7.9)
Random heights	1D	18	−33.9	−7.9	−17.8	−17.8	−31.0	9.2 (4.8..11.0)	13.5 (max 20.0)
	2D	18	−34.4	−14.8	−24.7	−24.7	−31.5	5.6 (0.2..10.8)	7.0 (max 13.3)
Cross-track ramp	1D	78	−38.7	−8.6	−9.7	−9.7	−38.9	1.2 (-12.5..19.3)	29.2 (max 46.0)
	2D	78	−39.2	−23.1	−38.3	−38.3	−39.3	11.9 (-5.9..25.6)	0.3 (max 25.5)
all cases	1D	259	−34.3	−12.1	−27.4	−27.4	−31.0	6.3 (-12.5..25.9)	3.9 (max 46.0)
	2D	259	−34.4	−24.6	−31.5	−31.5	−31.5	6.4 (-5.9..25.6)	0.1 (max 25.5)

Table 10: Worst ambiguity [dB], medians over the cases of each family, both algorithms. gain = no SATA − SATA whole band (median, and range over the family); gap = SATA whole band − oracle. Per sub-band differs from whole band by at most 0.07 dB (1D) and 0.07 dB (2D) over all cases. The 2-D metric is milder on "no SATA" than the 1-D one (Section 6); compare the gaps to the oracle.

9 Selected cases

The cases below are the same for both algorithms, one page per algorithm and case. The SATA 2D page adds the panels of Sakar's Fig. 4.20 for the representative target (the one the per-sub-band pass recovers worst): the 2-D impulse response after SATA per sub-band, contoured in dB relative to the reference peak (in the cross-track ramps the neighbours in range appear in the same panel at azimuth 0), and the phase of the reconstructed 2-D spectrum relative to the reference along range frequency at $f_a = 0$ and along azimuth frequency at $f_r = 0$, for no SATA, SATA per sub-band and the oracle. Where the error is comparable to the signal (no SATA, or the unresolved scenes) that phase is the phase of a superposition and reads as noise; where SATA works it is a few degrees, and where the oracle is not at zero the remainder is the reconstruction's own.

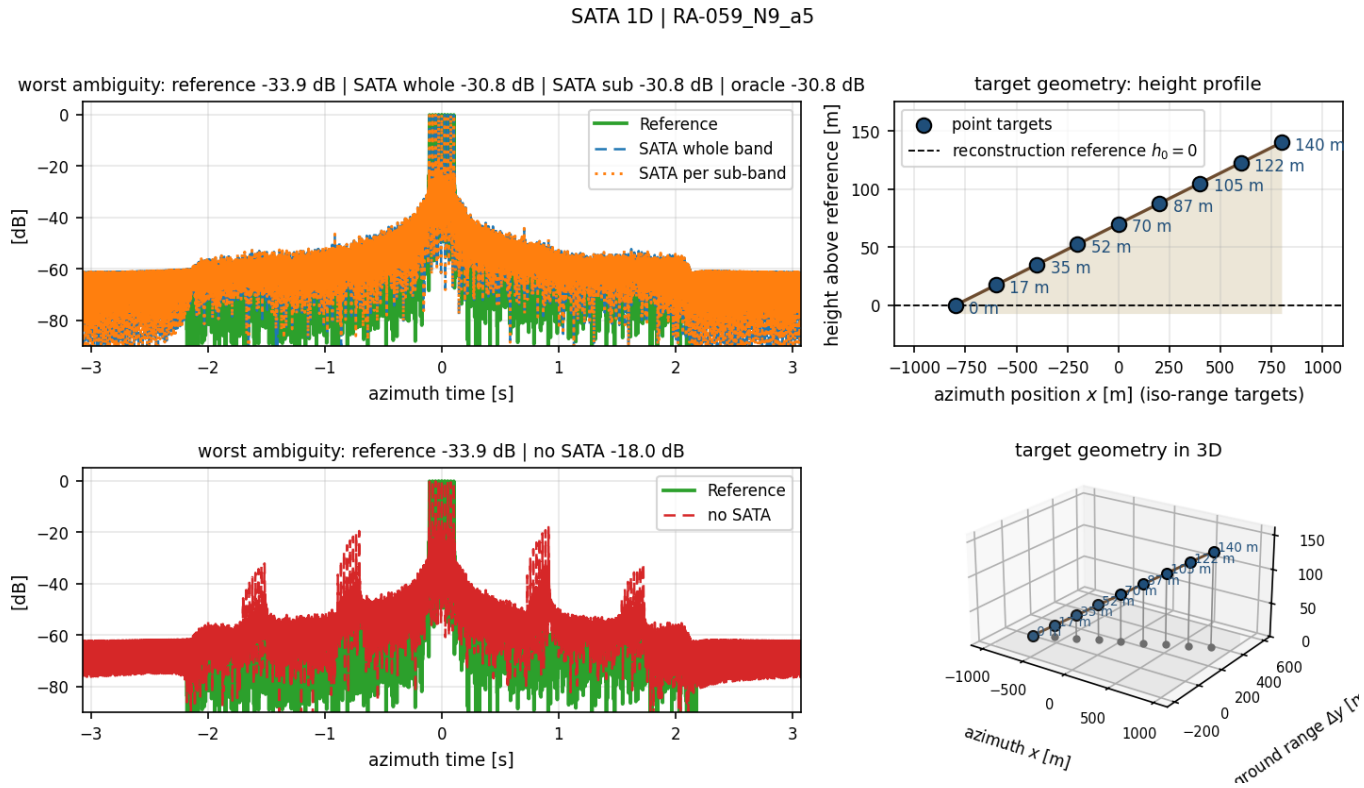


Figure 12: SATA 1D — Along-track ramp, case `N9_a5`: $N = 9$ iso-range targets at x = -800..800 m (200 m apart), heights 0, 17, 35, 52, 70, 87, 105, 122, 140 m. Worst ambiguity: reference −33.9, no SATA −18.0, SATA whole −30.8, SATA per sub-band −30.8, oracle −30.8 dB.

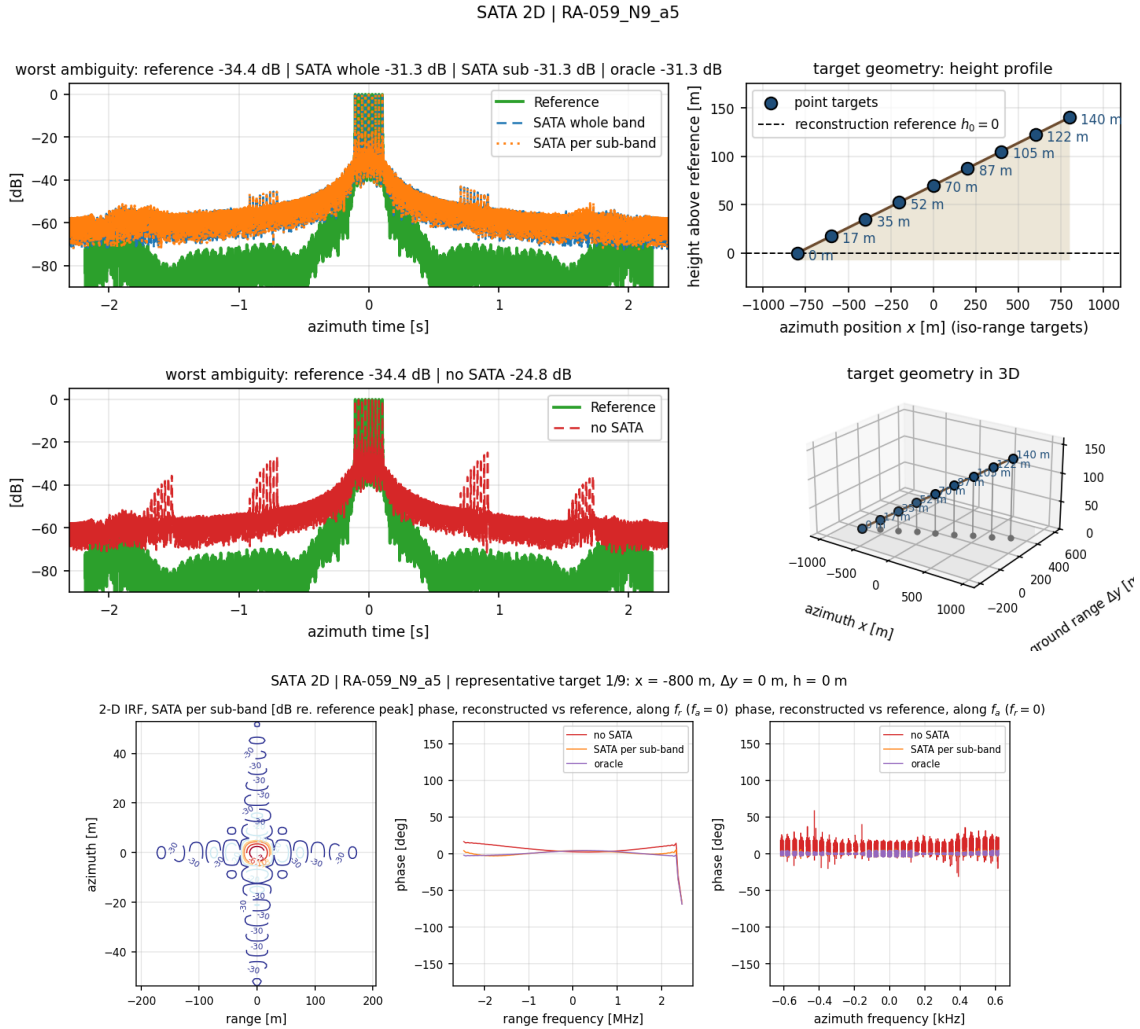


Figure 13: SATA 2D — Along-track ramp, case `N9_a5`: $N = 9$ iso-range targets at x = -800..800 m (200 m apart), heights 0, 17, 35, 52, 70, 87, 105, 122, 140 m. Worst ambiguity: reference −34.4, no SATA −24.8, SATA whole −31.3, SATA per sub-band −31.3, oracle −31.3 dB. Bottom row: 2-D IRF of the representative target after SATA per sub-band and the spectral phase error along f_r and f_a.

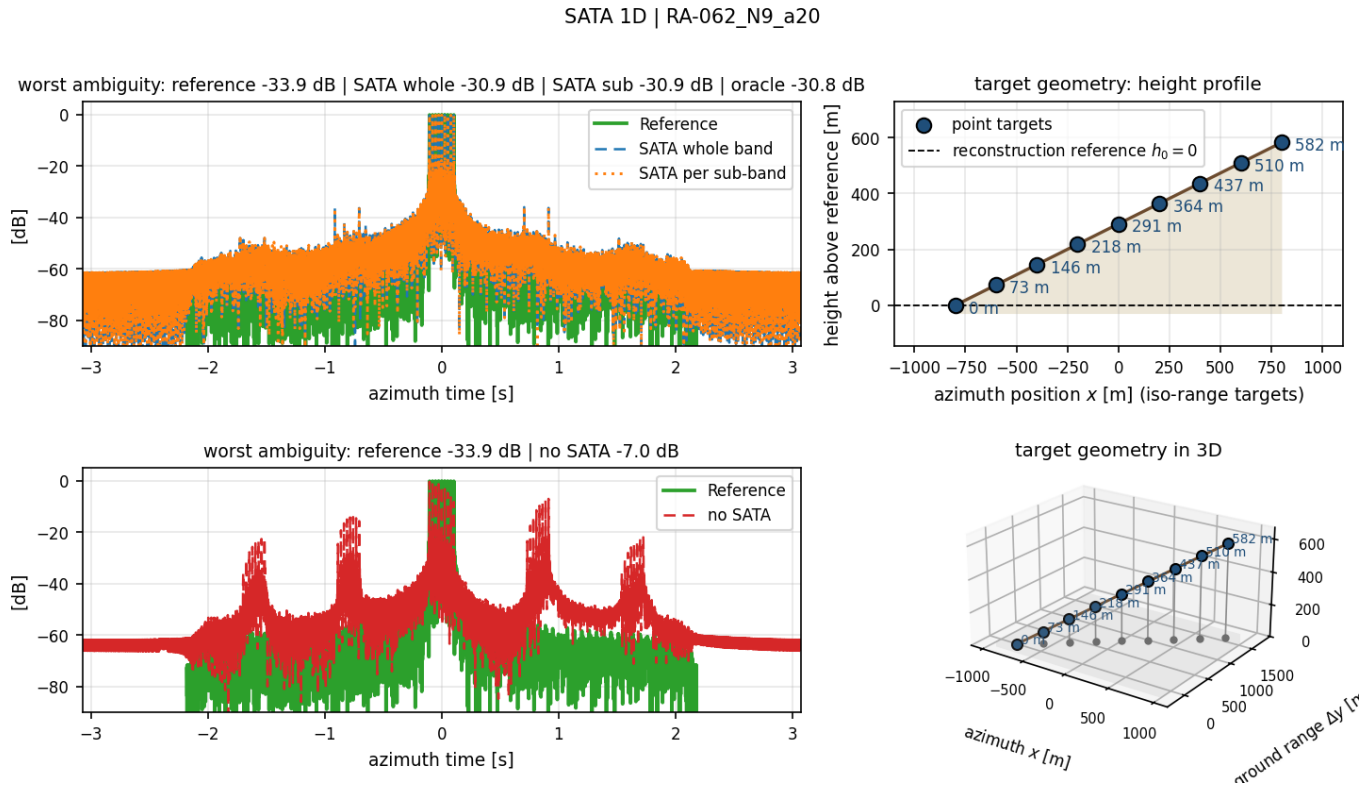


Figure 14: SATA 1D — Along-track ramp, case `N9_a20`: $N = 9$ iso-range targets at x = -800..800 m (200 m apart), heights 0, 73, 146, 218, 291, 364, 437, 510, 582 m. Worst ambiguity: reference −33.9, no SATA −7.0, SATA whole −30.9, SATA per sub-band −30.9, oracle −30.8 dB.

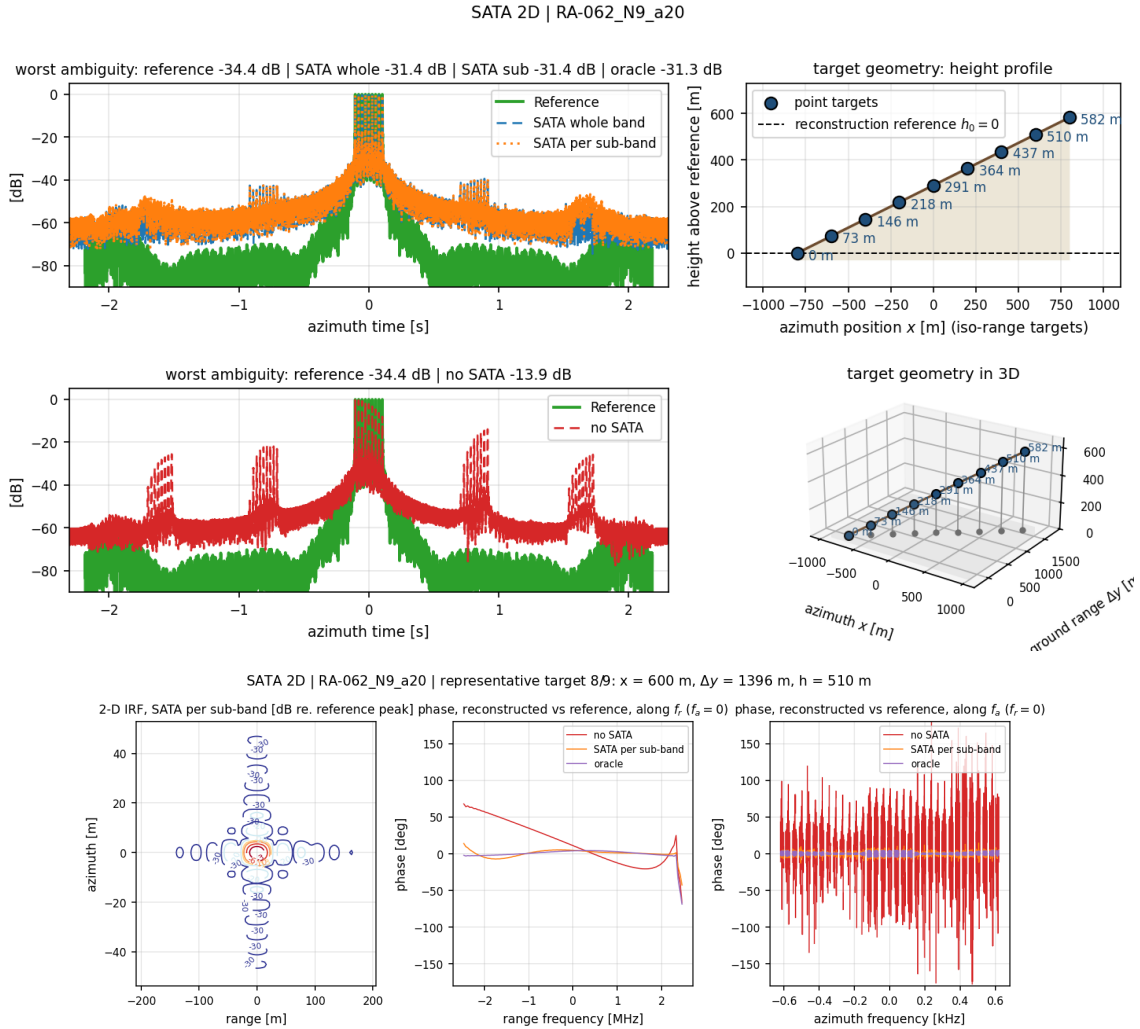


Figure 15: SATA 2D — Along-track ramp, case `N9_a20`: $N = 9$ iso-range targets at x = -800..800 m (200 m apart), heights 0, 73, 146, 218, 291, 364, 437, 510, 582 m. Worst ambiguity: reference −34.4, no SATA −13.9, SATA whole −31.4, SATA per sub-band −31.4, oracle −31.3 dB. Bottom row: 2-D IRF of the representative target after SATA per sub-band and the spectral phase error along f_r and f_a.

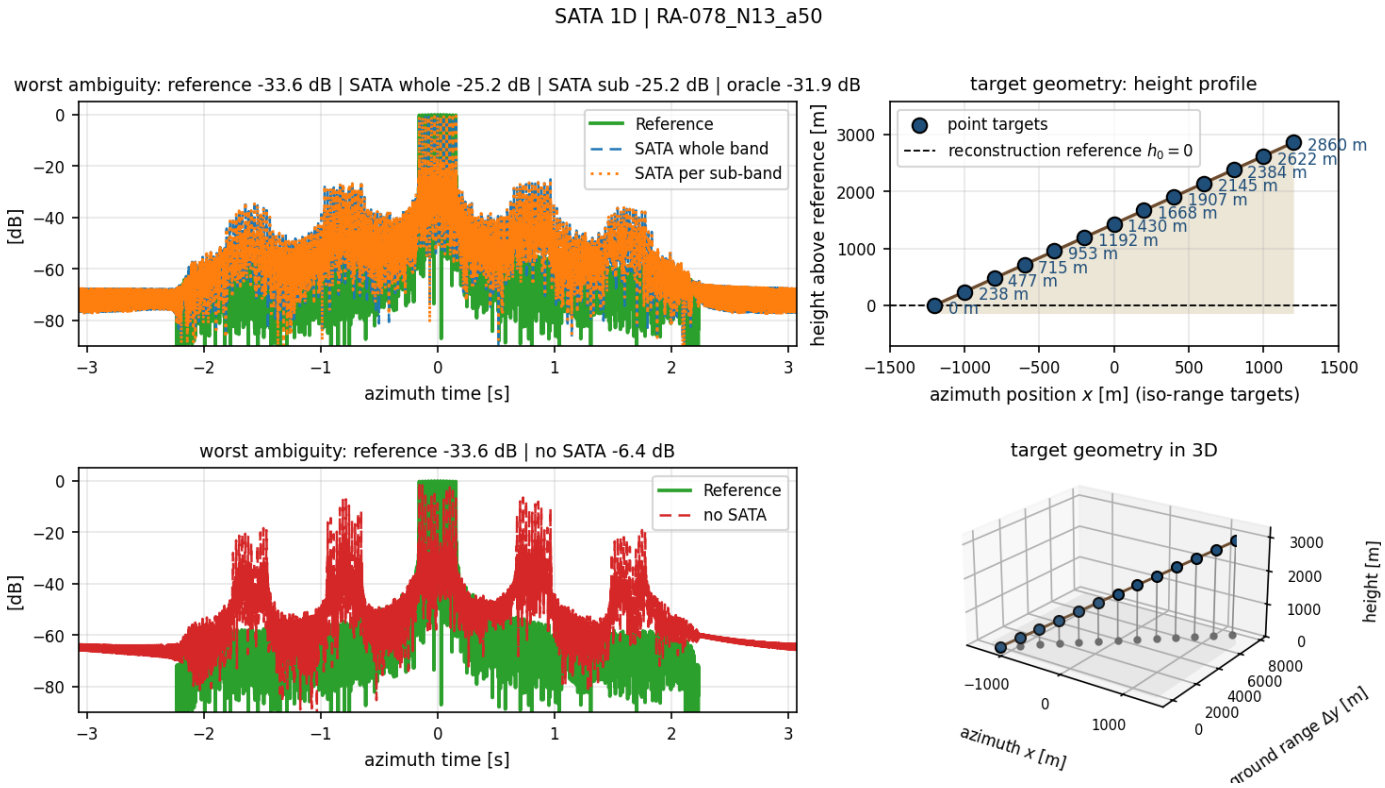


Figure 16: SATA 1D — Along-track ramp, case `N13_a50`: $N = 13$ iso-range targets at x = -1200..1200 m (200 m apart), heights 0, 238, 477, 715, 953, 1192, 1430, 1668, 1907, 2145, 2384, 2622, 2860 m. Worst ambiguity: reference −33.6, no SATA −6.4, SATA whole −25.2, SATA per sub-band −25.2, oracle −31.9 dB.

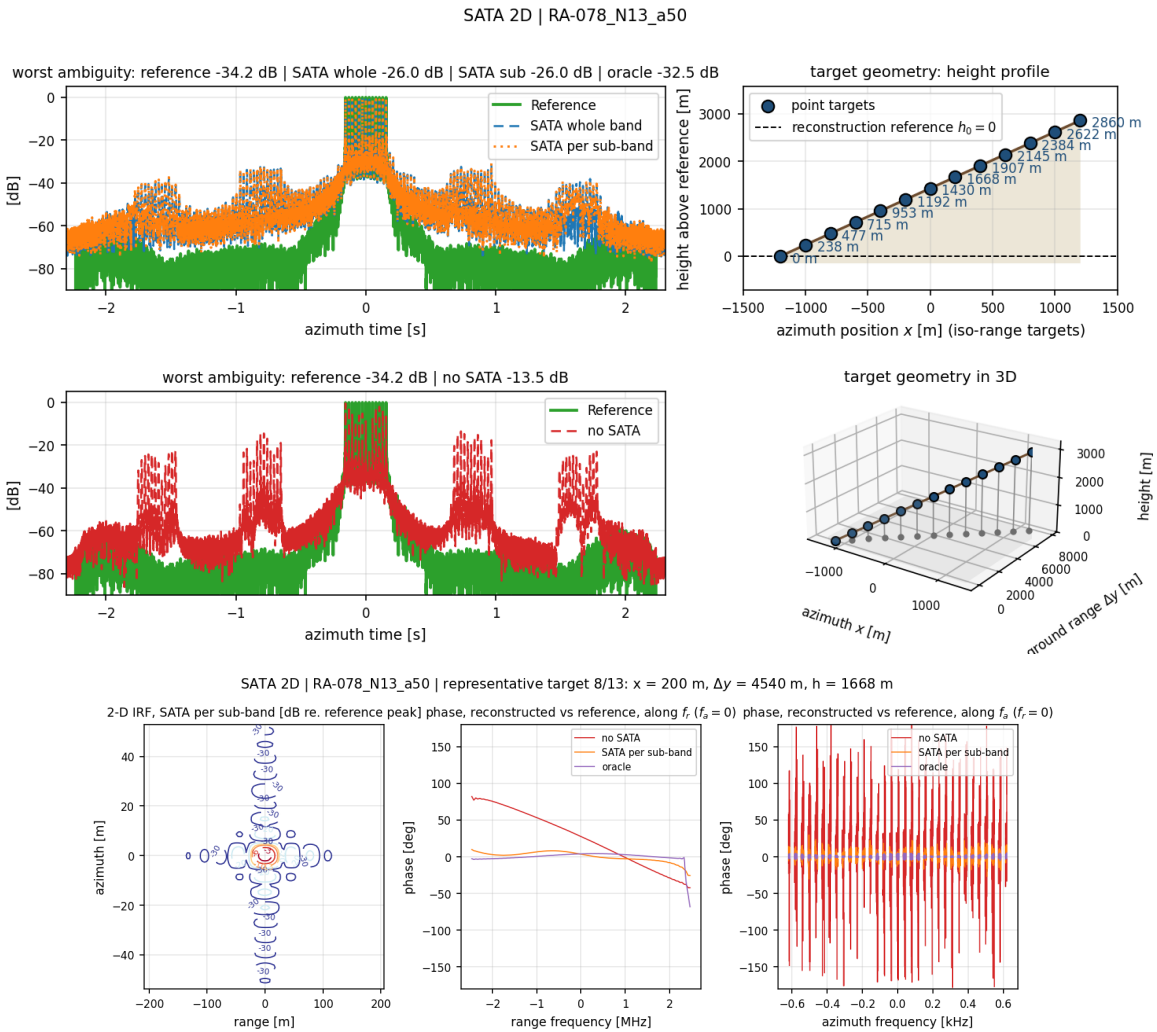


Figure 17: SATA 2D — Along-track ramp, case `N13_a50`: $N = 13$ iso-range targets at x = -1200..1200 m (200 m apart), heights 0, 238, 477, 715, 953, 1192, 1430, 1668, 1907, 2145, 2384, 2622, 2860 m. Worst ambiguity: reference −34.2, no SATA −13.5, SATA whole −26.0, SATA per sub-band −26.0, oracle −32.5 dB. Bottom row: 2-D IRF of the representative target after SATA per sub-band and the spectral phase error along f_r and f_a.

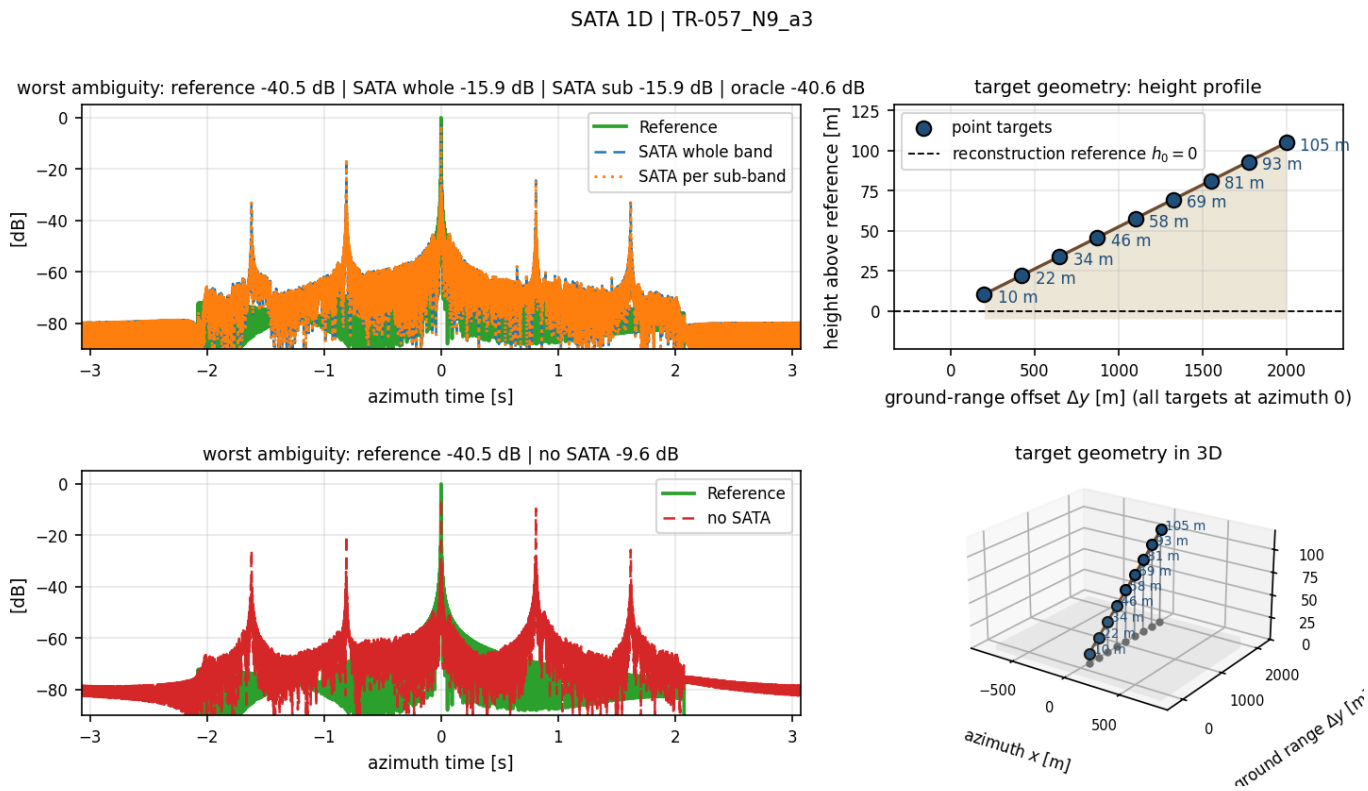


Figure 18: SATA 1D — Cross-track ramp, case `N9_a3`: $N = 9$ targets at azimuth 0, $\Delta y = 200..2000$ m, heights 10..105 m. Worst ambiguity: reference −40.5, no SATA −9.6, SATA whole −15.9, SATA per sub-band −15.9, oracle −40.6 dB.

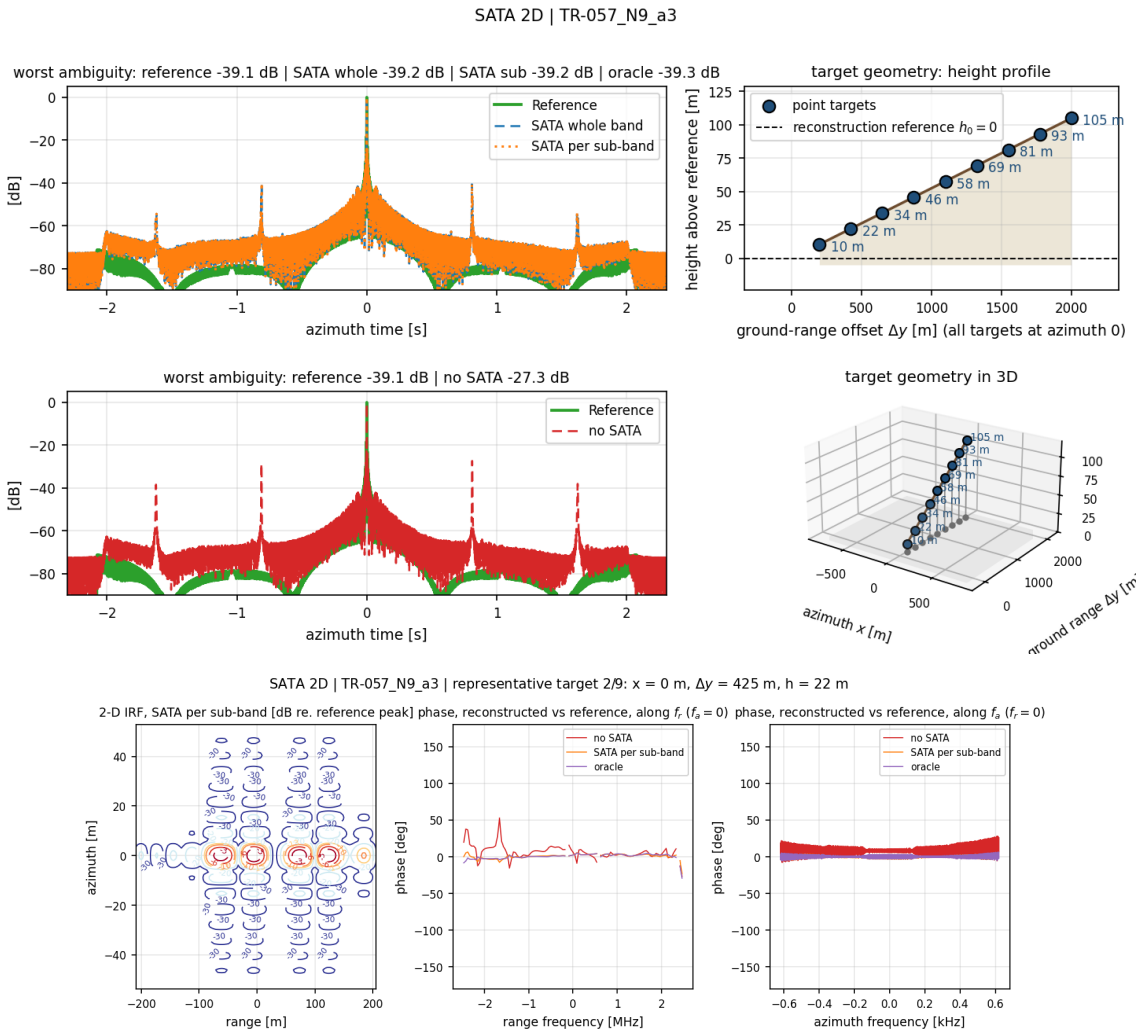


Figure 19: SATA 2D — Cross-track ramp, case `N9_a3`: $N = 9$ targets at azimuth 0, $\Delta y = 200..2000$ m, heights 10..105 m. Worst ambiguity: reference −39.1, no SATA −27.3, SATA whole −39.2, SATA per sub-band −39.2, oracle −39.3 dB. Bottom row: 2-D IRF of the representative target after SATA per sub-band and the spectral phase error along f_r and f_a.

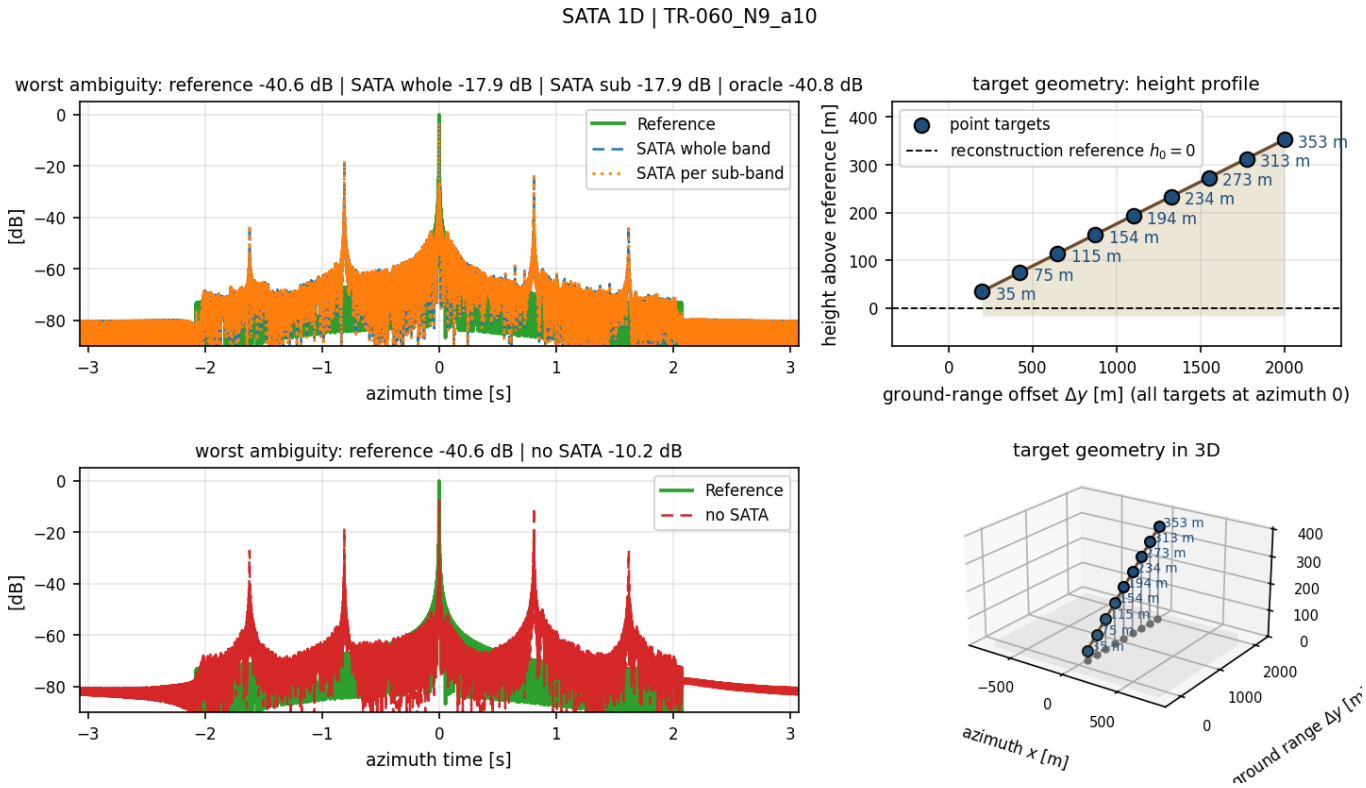


Figure 20: SATA 1D — Cross-track ramp, case `N9_a10`: $N = 9$ targets at azimuth 0, $\Delta y = 200..2000$ m, heights 35..353 m. Worst ambiguity: reference −40.6, no SATA −10.2, SATA whole −17.9, SATA per sub-band −17.9, oracle −40.8 dB.

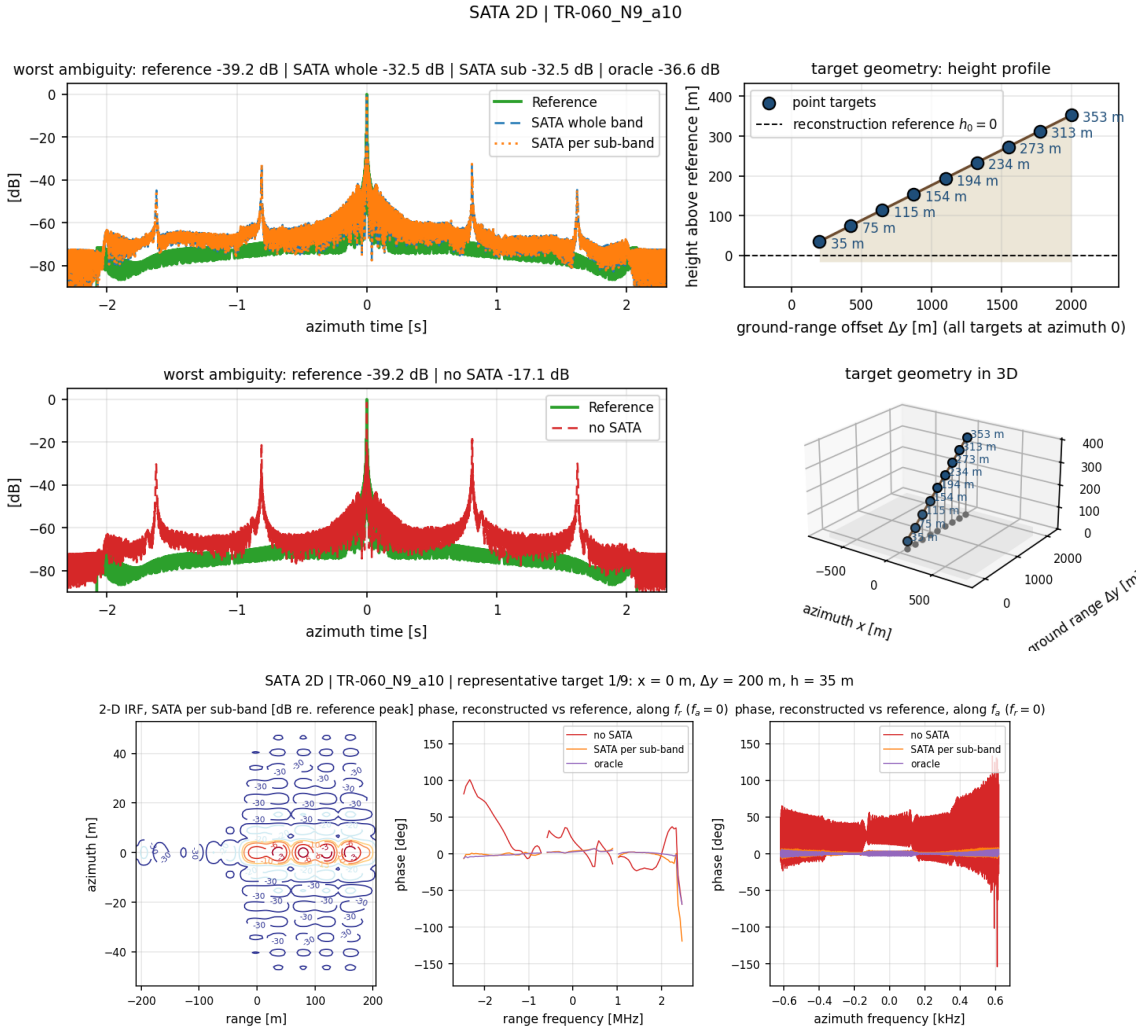


Figure 21: SATA 2D — Cross-track ramp, case `N9_a10`: $N = 9$ targets at azimuth 0, $\Delta y = 200..2000$ m, heights 35..353 m. Worst ambiguity: reference −39.2, no SATA −17.1, SATA whole −32.5, SATA per sub-band −32.5, oracle −36.6 dB. Bottom row: 2-D IRF of the representative target after SATA per sub-band and the spectral phase error along f_r and f_a.

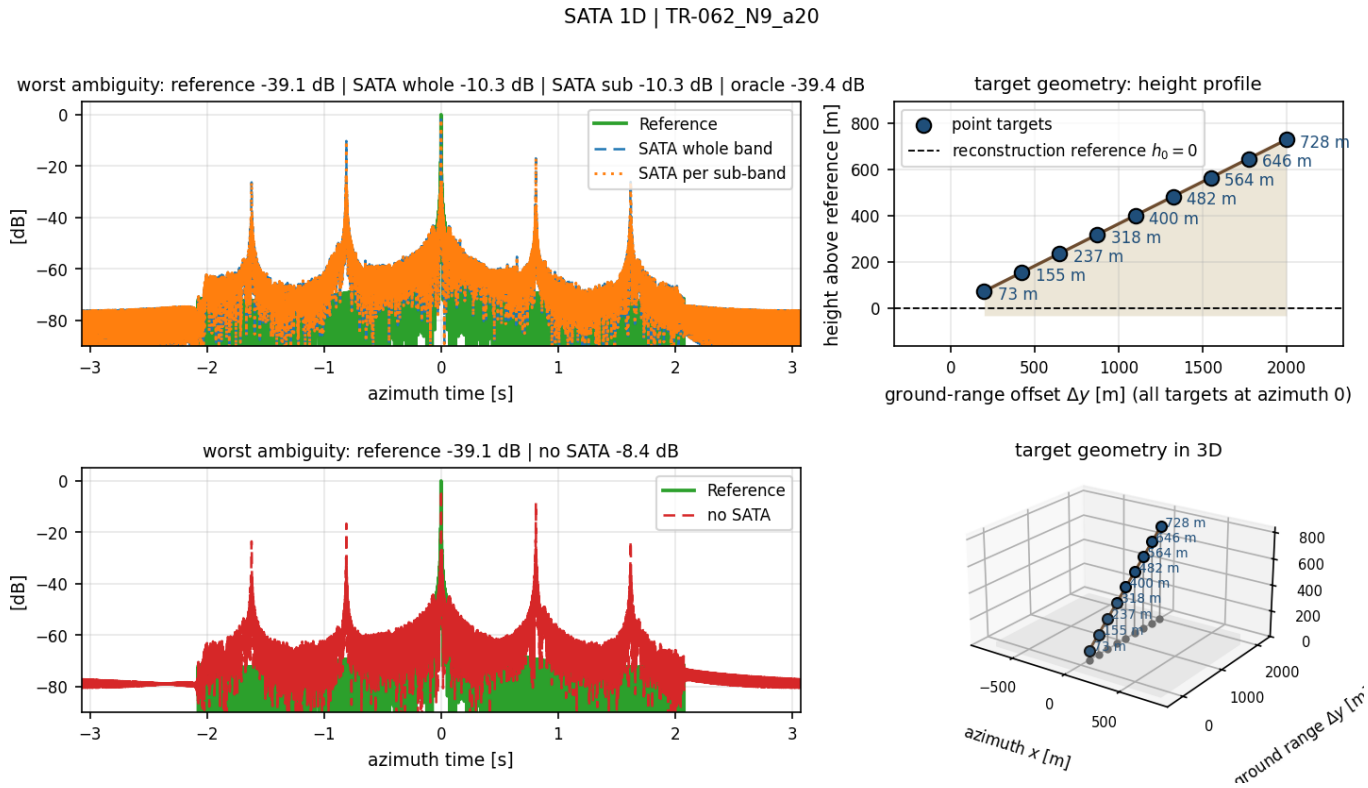


Figure 22: SATA 1D — Cross-track ramp, case `N9_a20`: $N = 9$ targets at azimuth 0, $\Delta y = 200..2000$ m, heights 73..728 m. Worst ambiguity: reference −39.1, no SATA −8.4, SATA whole −10.3, SATA per sub-band −10.3, oracle −39.4 dB.

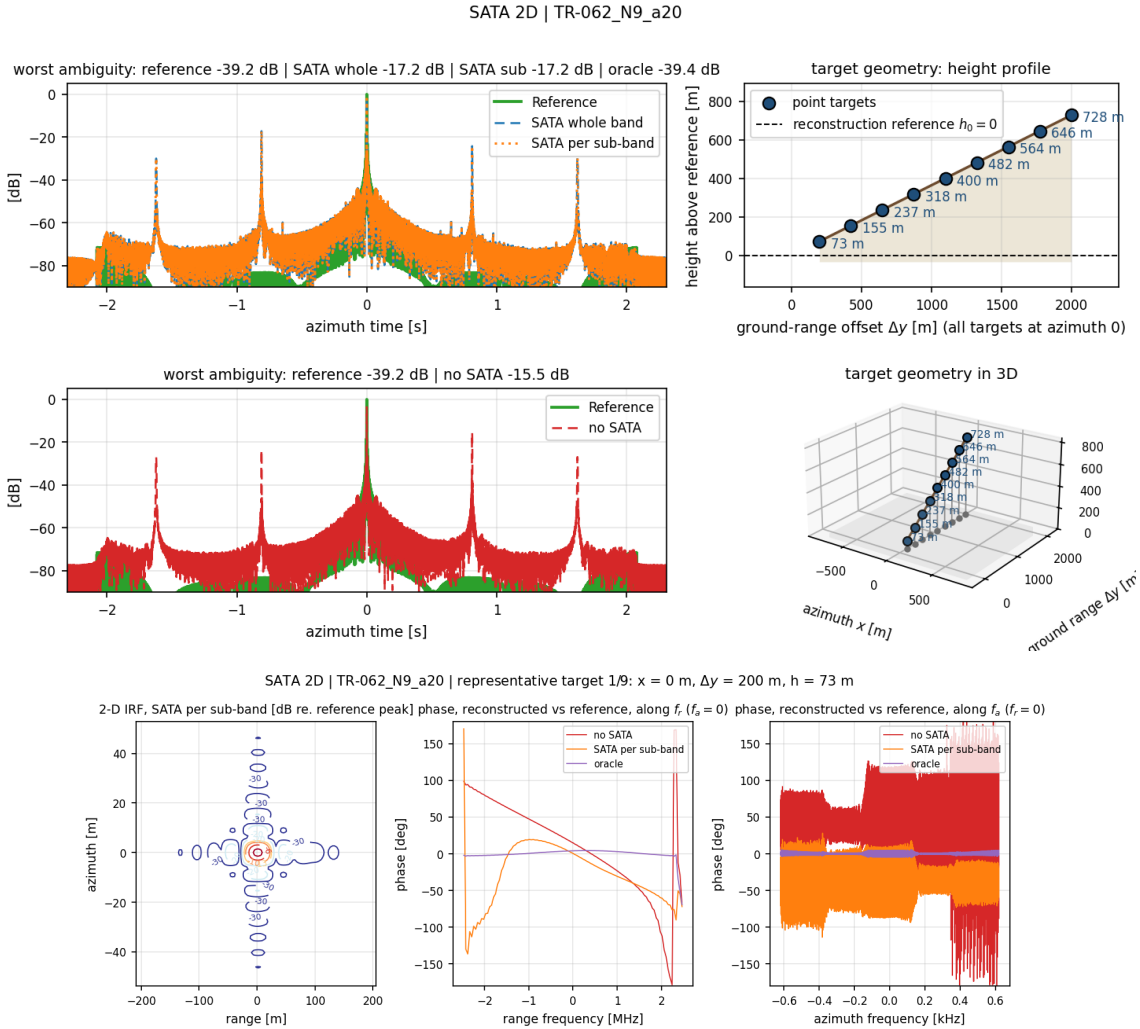


Figure 23: SATA 2D — Cross-track ramp, case `N9_a20`: $N = 9$ targets at azimuth 0, $\Delta y = 200..2000$ m, heights 73..728 m. Worst ambiguity: reference −39.2, no SATA −15.5, SATA whole −17.2, SATA per sub-band −17.2, oracle −39.4 dB. Bottom row: 2-D IRF of the representative target after SATA per sub-band and the spectral phase error along f_r and f_a.

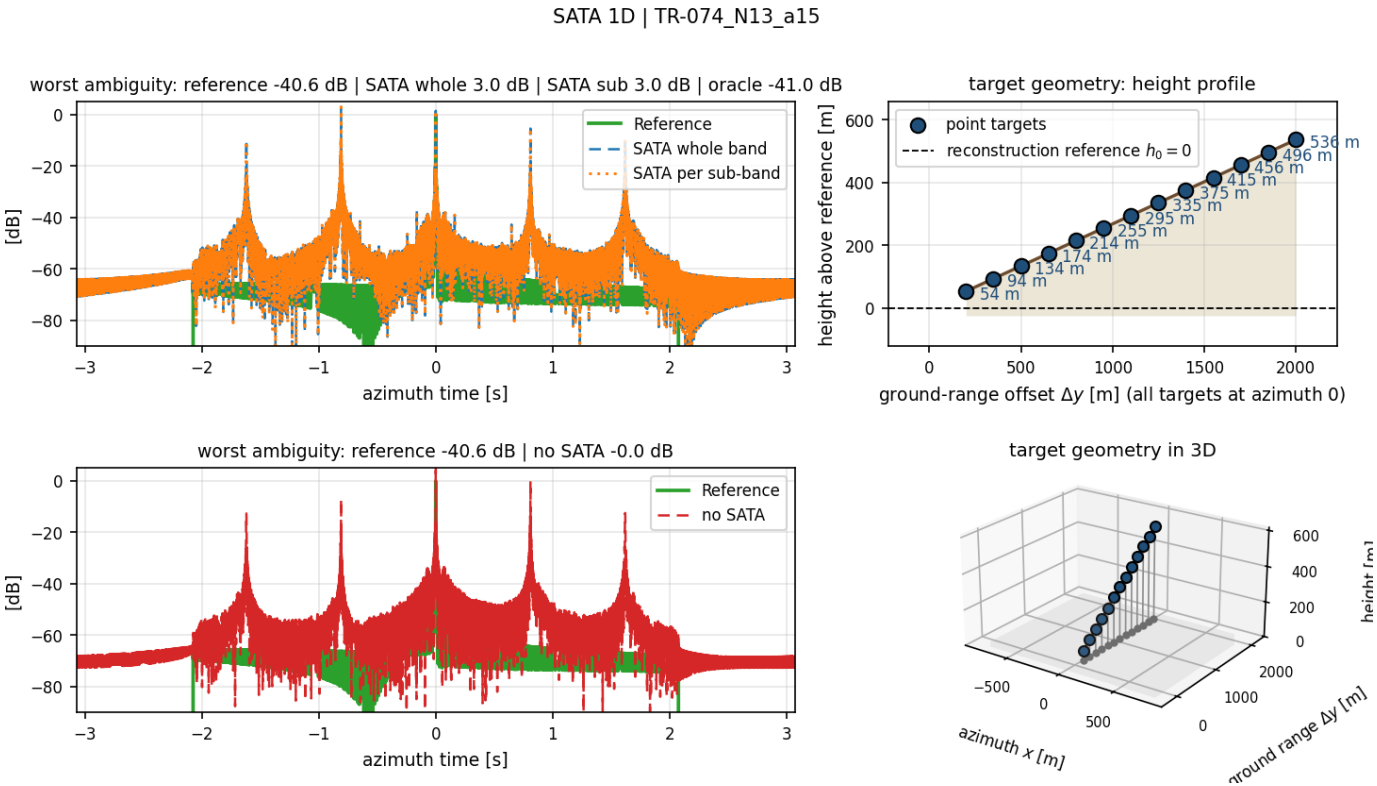


Figure 24: SATA 1D — Cross-track ramp, case `N13_a15`: $N = 13$ targets at azimuth 0, $\Delta y = 200..2000$ m, heights 54..536 m. Worst ambiguity: reference −40.6, no SATA −0.0, SATA whole 3.0, SATA per sub-band 3.0, oracle −41.0 dB.

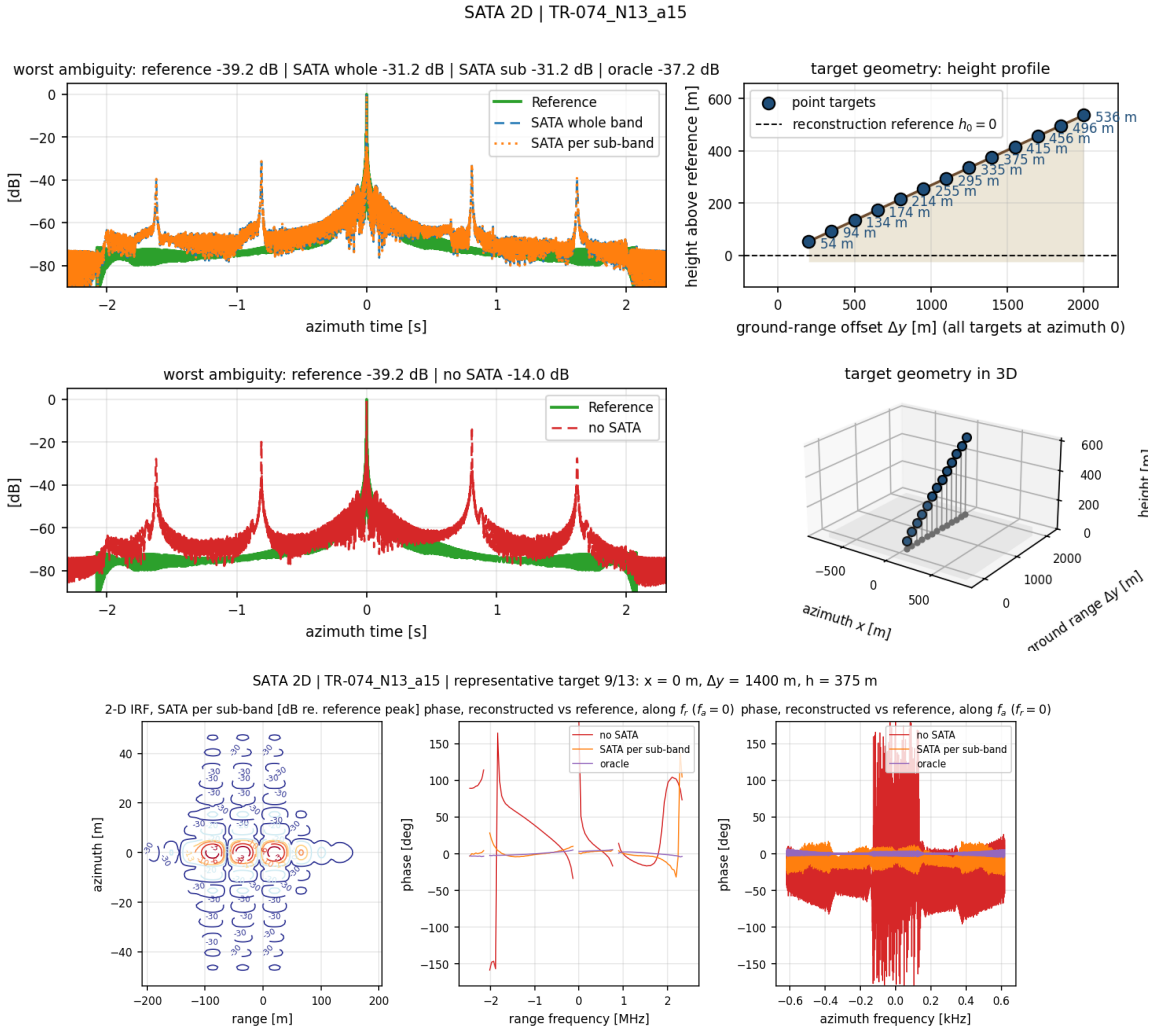


Figure 25: SATA 2D — Cross-track ramp, case `N13_a15`: $N = 13$ targets at azimuth 0, $\Delta y = 200..2000$ m, heights 54..536 m. Worst ambiguity: reference −39.2, no SATA −14.0, SATA whole −31.2, SATA per sub-band −31.2, oracle −37.2 dB. Bottom row: 2-D IRF of the representative target after SATA per sub-band and the spectral phase error along f_r and f_a.

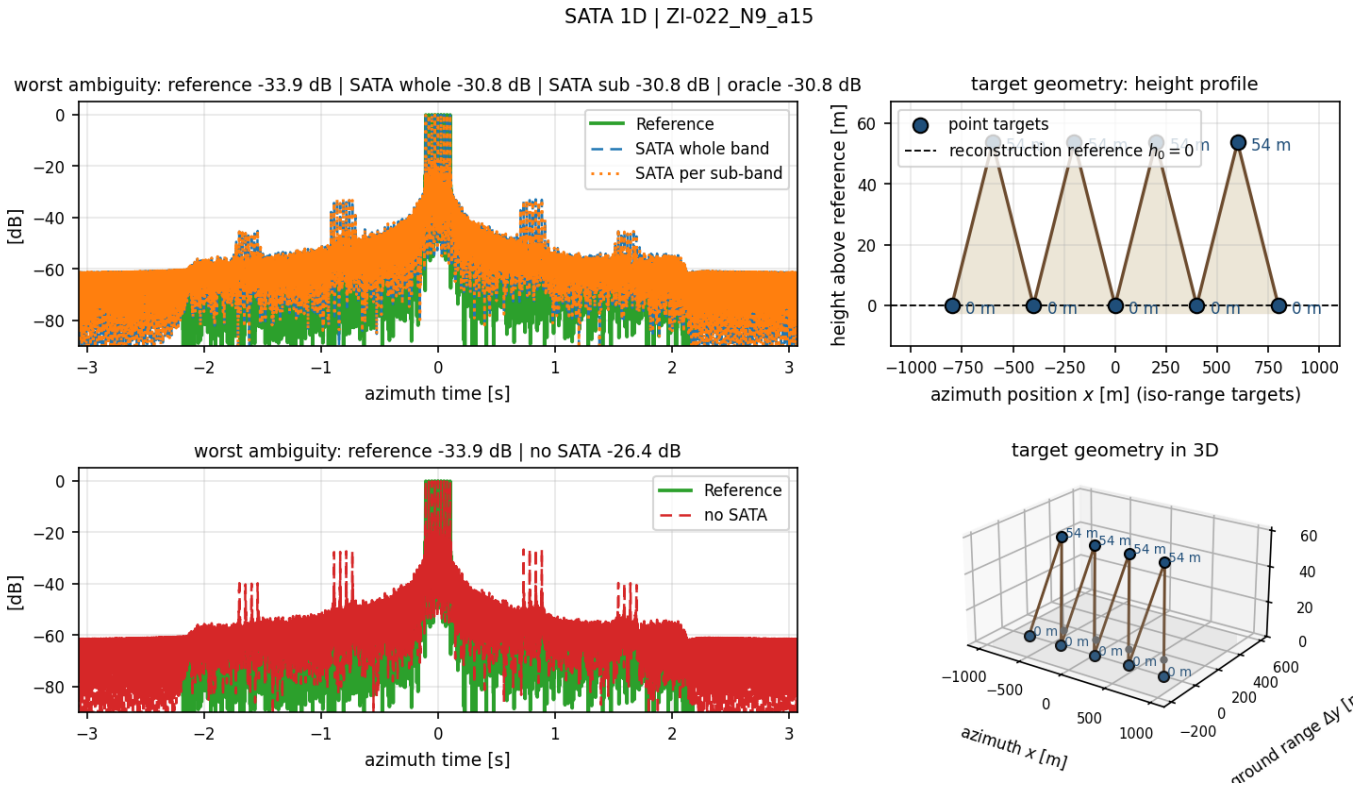


Figure 26: SATA 1D — Zig-zag ramp, case `N9_a15`: $N = 9$ iso-range targets at x = -800..800 m (200 m apart), heights 0, 54, 0, 54, 0, 54, 0, 54, 0 m. Worst ambiguity: reference −33.9, no SATA −26.4, SATA whole −30.8, SATA per sub-band −30.8, oracle −30.8 dB.

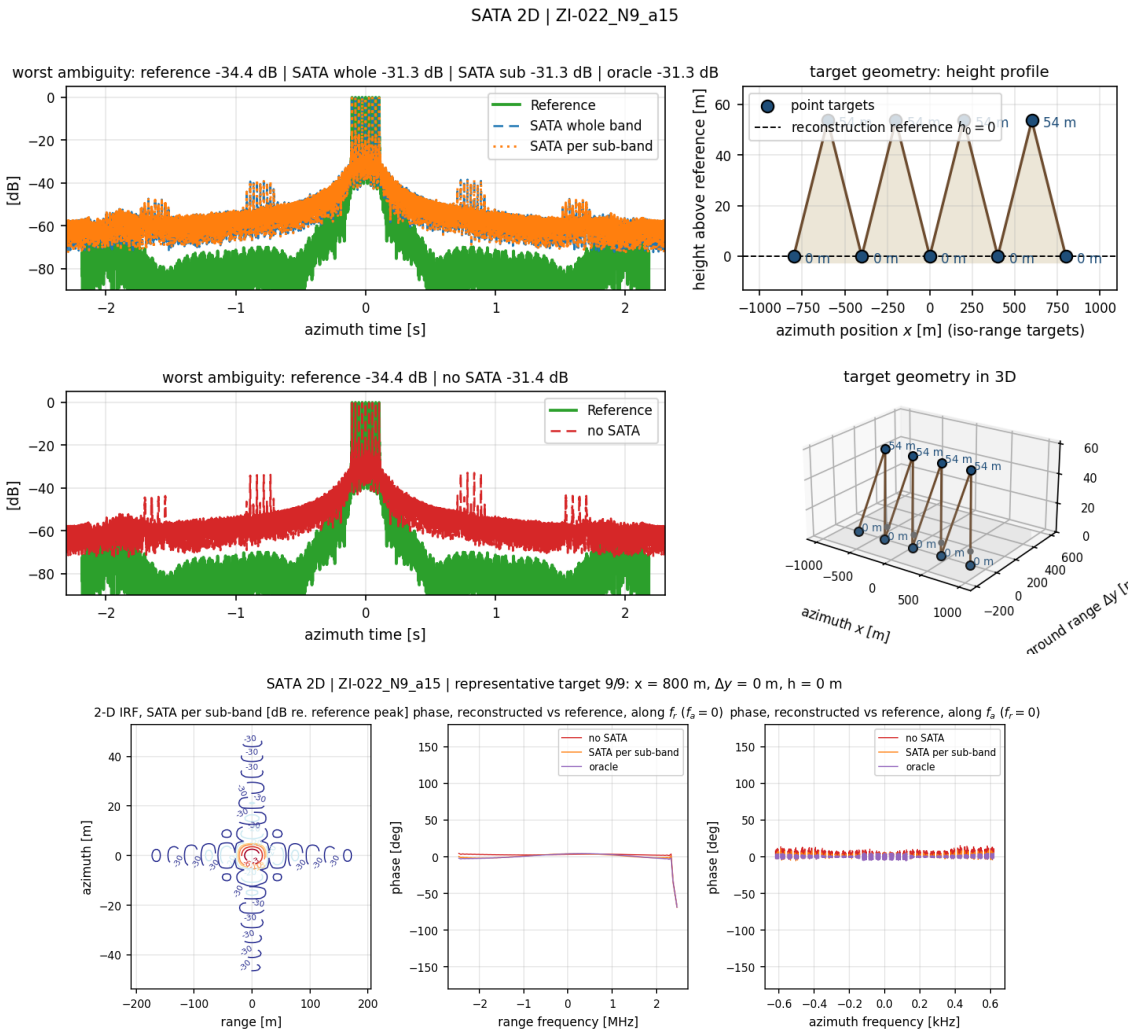


Figure 27: SATA 2D — Zig-zag ramp, case `N9_a15`: $N = 9$ iso-range targets at x = -800..800 m (200 m apart), heights 0, 54, 0, 54, 0, 54, 0, 54, 0 m. Worst ambiguity: reference −34.4, no SATA −31.4, SATA whole −31.3, SATA per sub-band −31.3, oracle −31.3 dB. Bottom row: 2-D IRF of the representative target after SATA per sub-band and the spectral phase error along f_r and f_a.

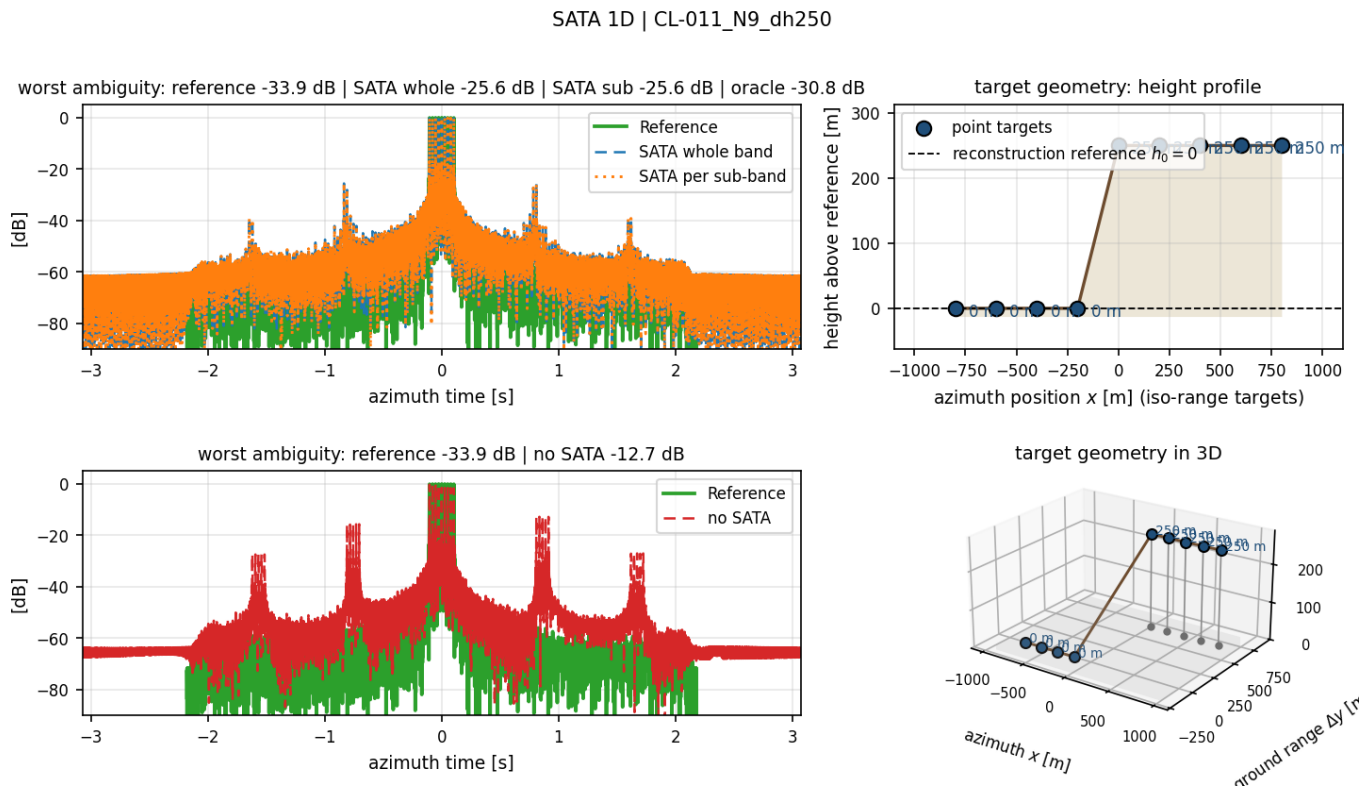


Figure 28: SATA 1D — Cliff, case `N9_dh250`: $N = 9$ iso-range targets at x = -800..800 m (200 m apart), heights 0, 0, 0, 0, 250, 250, 250, 250, 250 m. Worst ambiguity: reference −33.9, no SATA −12.7, SATA whole −25.6, SATA per sub-band −25.6, oracle −30.8 dB.

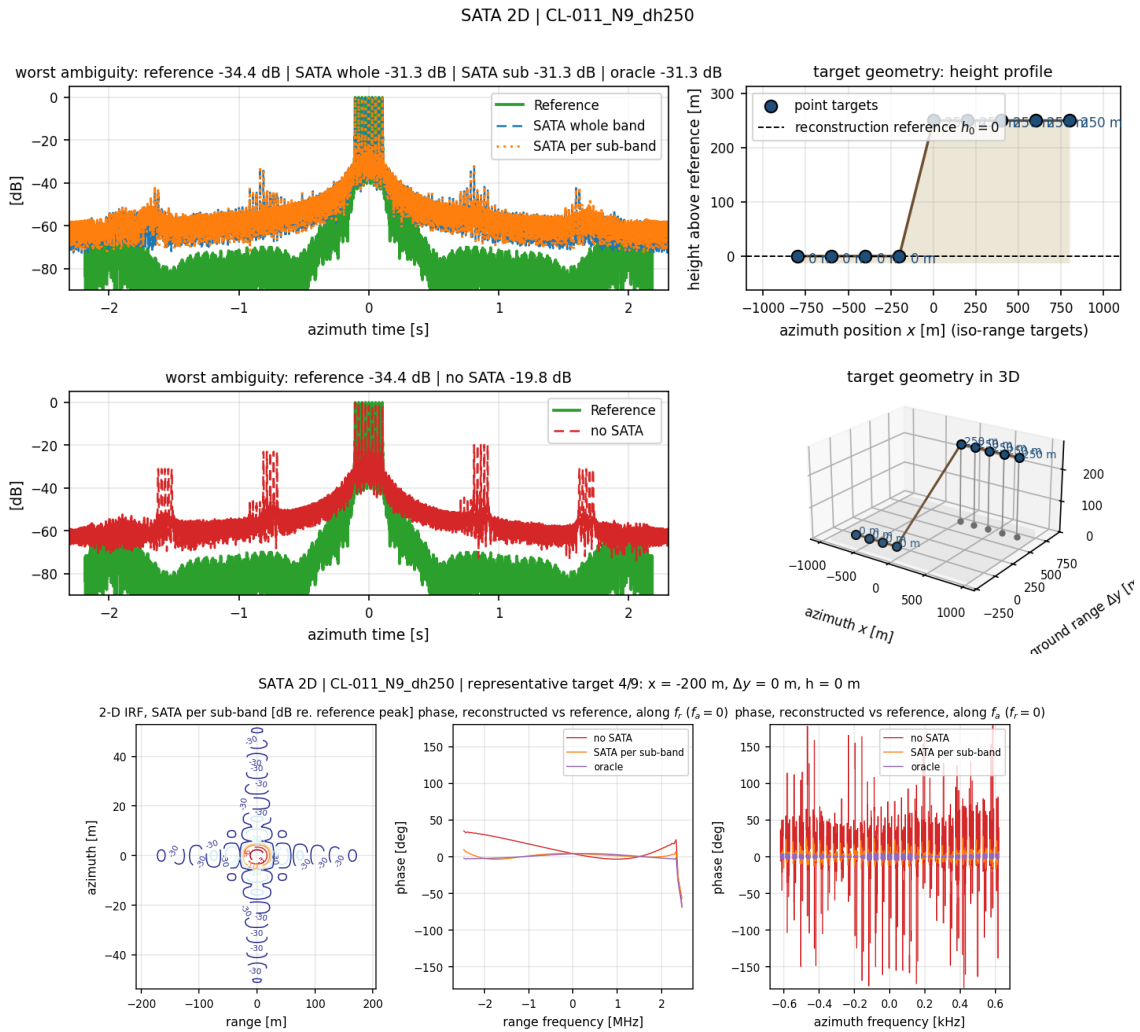


Figure 29: SATA 2D — Cliff, case `N9_dh250`: $N = 9$ iso-range targets at x = -800..800 m (200 m apart), heights 0, 0, 0, 0, 250, 250, 250, 250, 250 m. Worst ambiguity: reference −34.4, no SATA −19.8, SATA whole −31.3, SATA per sub-band −31.3, oracle −31.3 dB. Bottom row: 2-D IRF of the representative target after SATA per sub-band and the spectral phase error along f_r and f_a.

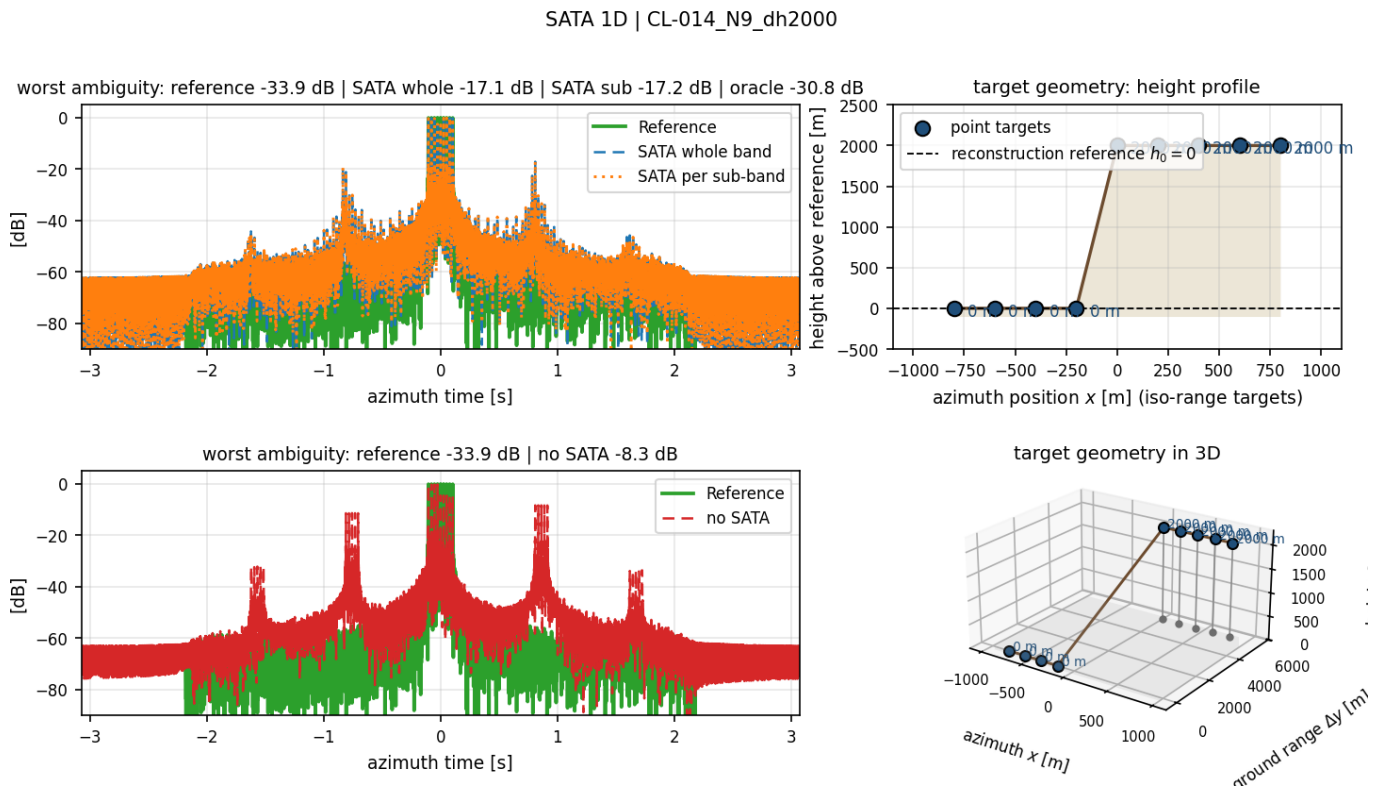


Figure 30: SATA 1D — Cliff, case `N9_dh2000`: $N = 9$ iso-range targets at x = -800..800 m (200 m apart), heights 0, 0, 0, 0, 2000, 2000, 2000, 2000, 2000 m. Worst ambiguity: reference −33.9, no SATA −8.3, SATA whole −17.1, SATA per sub-band −17.2, oracle −30.8 dB.

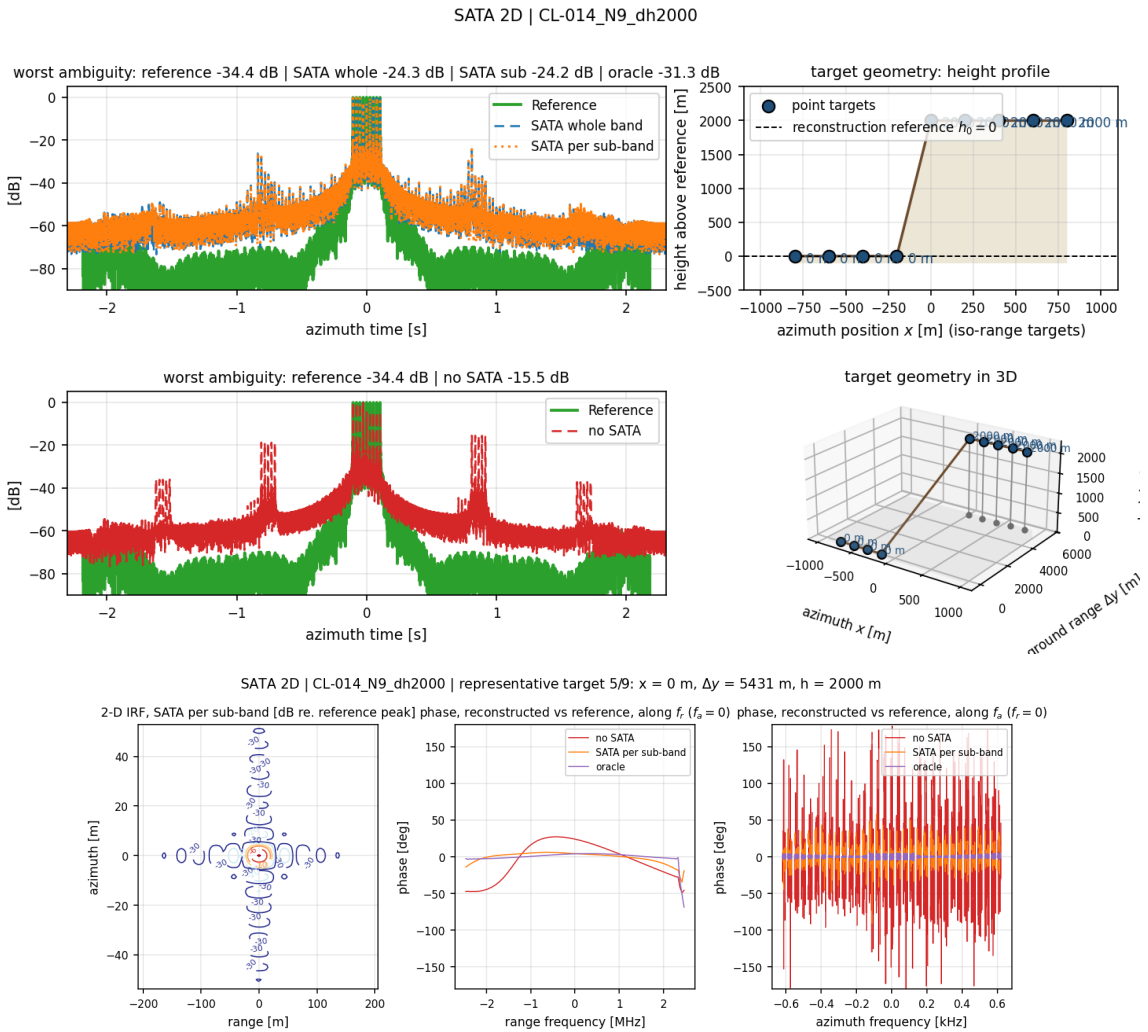


Figure 31: SATA 2D — Cliff, case `N9_dh2000`: $N = 9$ iso-range targets at x = -800..800 m (200 m apart), heights 0, 0, 0, 0, 2000, 2000, 2000, 2000, 2000 m. Worst ambiguity: reference −34.4, no SATA −15.5, SATA whole −24.3, SATA per sub-band −24.2, oracle −31.3 dB. Bottom row: 2-D IRF of the representative target after SATA per sub-band and the spectral phase error along f_r and f_a.

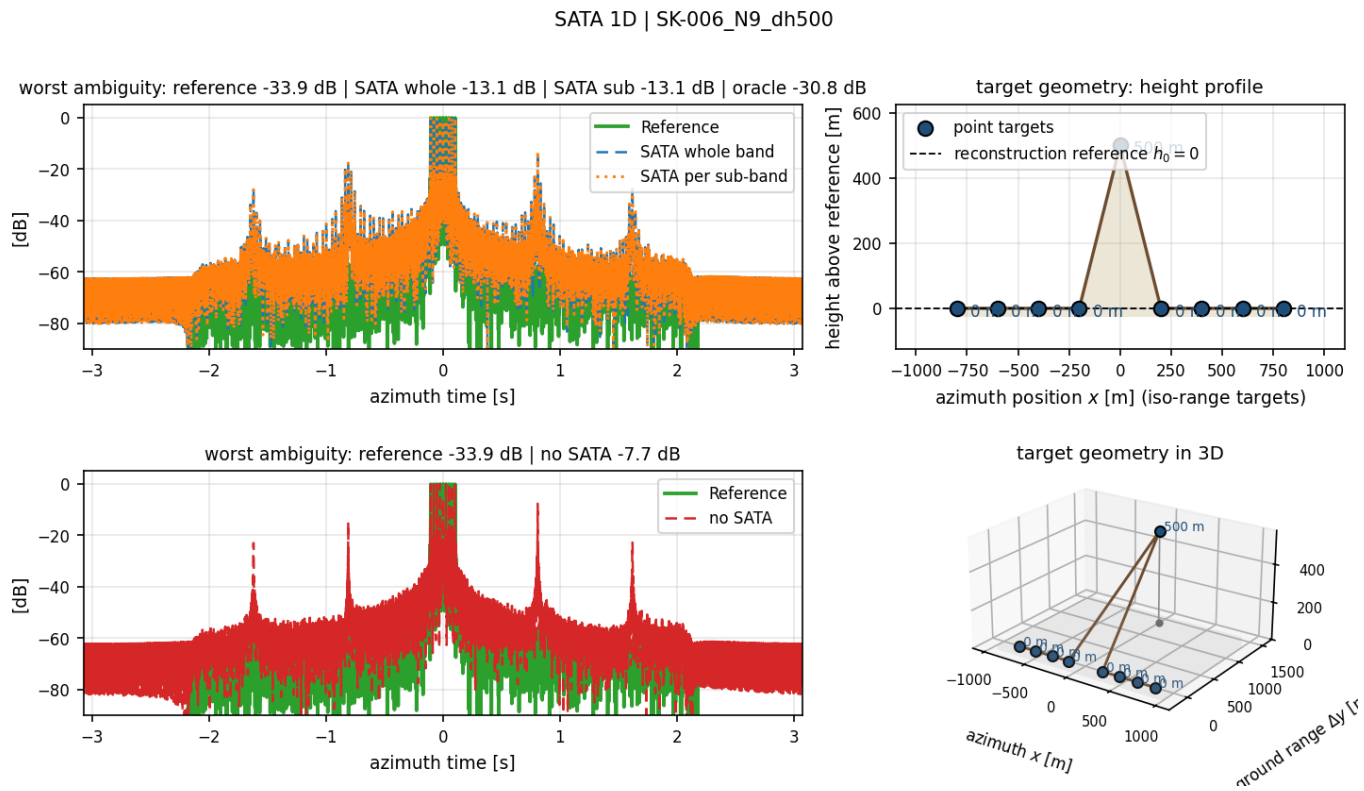


Figure 32: SATA 1D — Spike, case `N9_dh500`: $N = 9$ iso-range targets at x = -800..800 m (200 m apart), heights 0, 0, 0, 0, 500, 0, 0, 0, 0 m. Worst ambiguity: reference −33.9, no SATA −7.7, SATA whole −13.1, SATA per sub-band −13.1, oracle −30.8 dB.

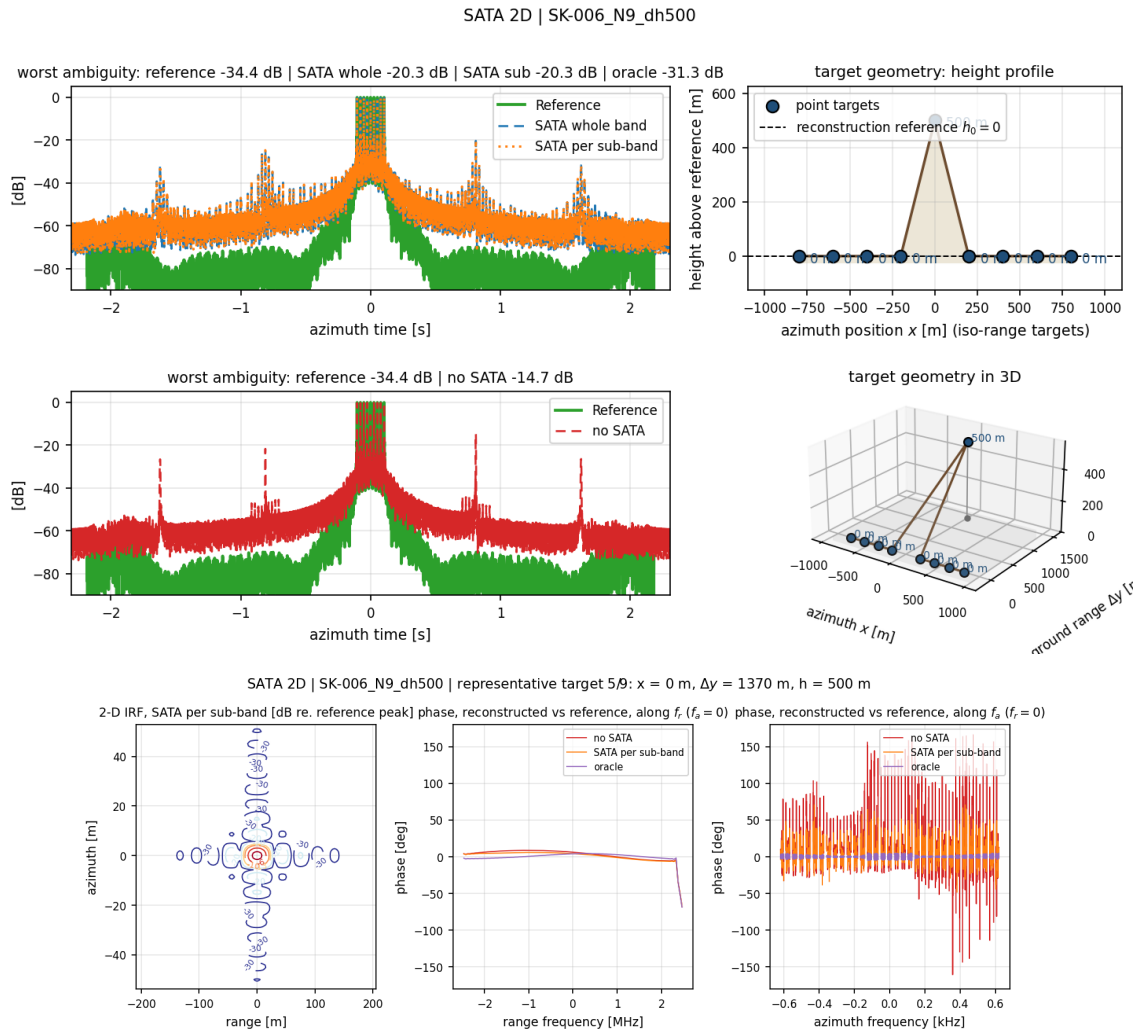


Figure 33: SATA 2D — Spike, case `N9_dh500`: $N = 9$ iso-range targets at x = -800..800 m (200 m apart), heights 0, 0, 0, 0, 500, 0, 0, 0, 0 m. Worst ambiguity: reference −34.4, no SATA −14.7, SATA whole −20.3, SATA per sub-band −20.3, oracle −31.3 dB. Bottom row: 2-D IRF of the representative target after SATA per sub-band and the spectral phase error along f_r and f_a.

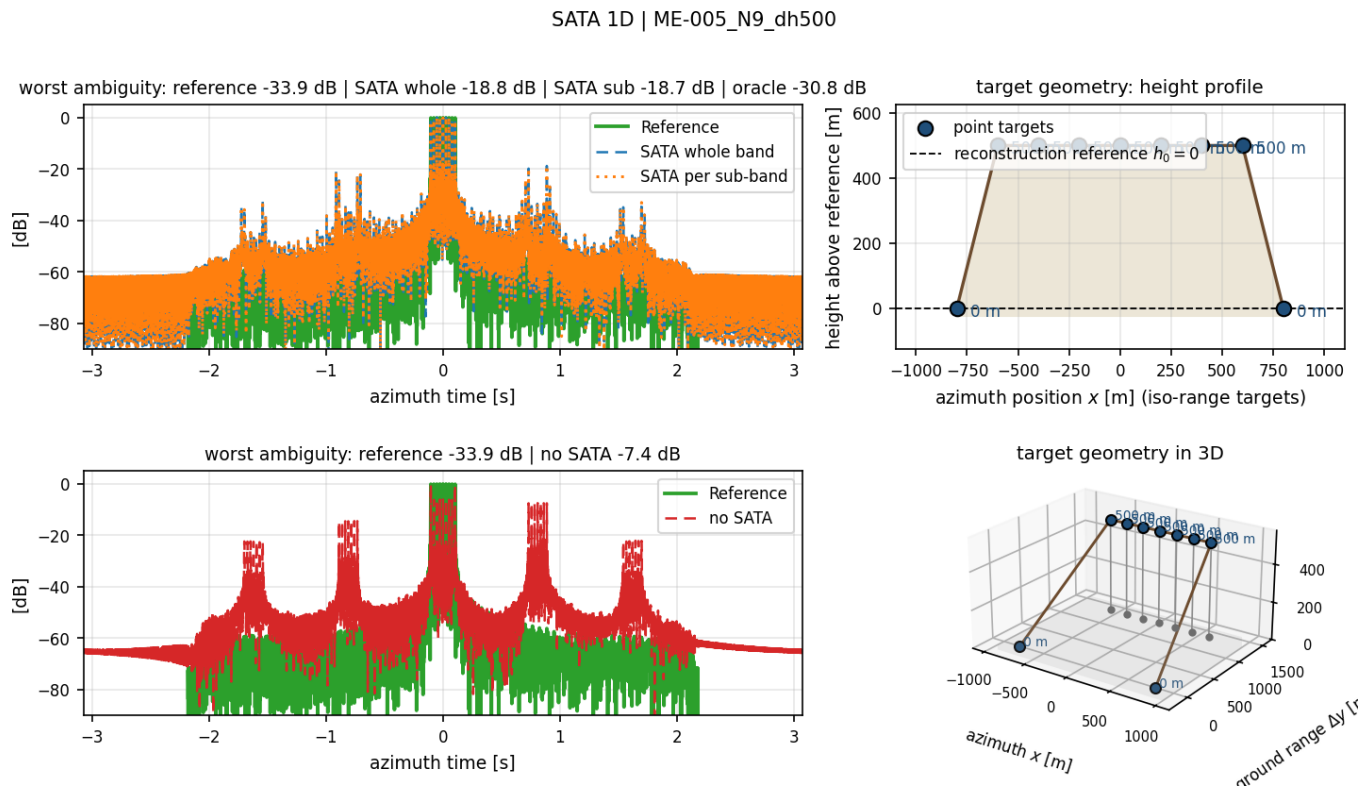


Figure 34: SATA 1D — Mesa, case `N9_dh500`: $N = 9$ iso-range targets at x = -800..800 m (200 m apart), heights 0, 500, 500, 500, 500, 500, 500, 500, 0 m. Worst ambiguity: reference −33.9, no SATA −7.4, SATA whole −18.8, SATA per sub-band −18.7, oracle −30.8 dB.

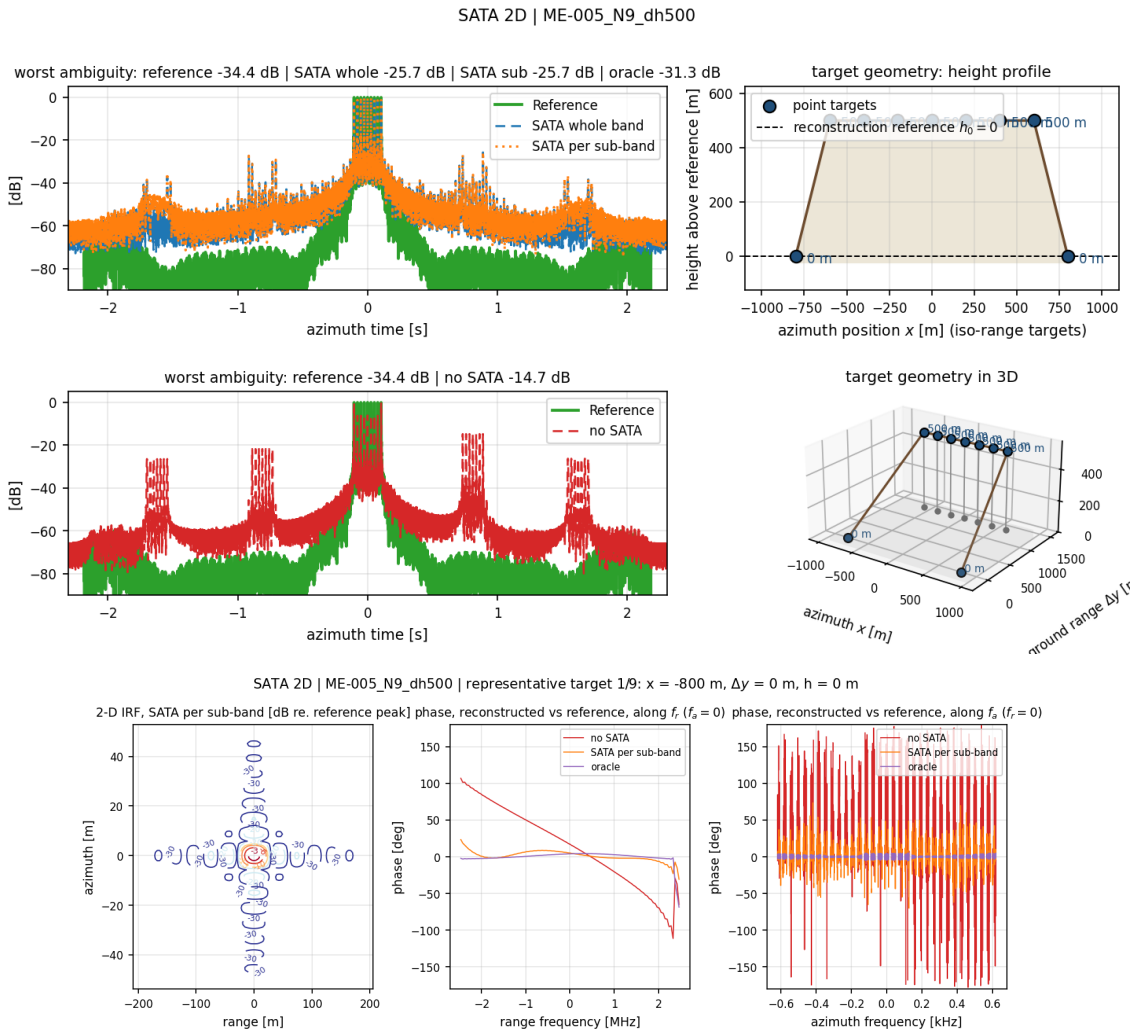


Figure 35: SATA 2D — Mesa, case `N9_dh500`: $N = 9$ iso-range targets at x = -800..800 m (200 m apart), heights 0, 500, 500, 500, 500, 500, 500, 500, 0 m. Worst ambiguity: reference −34.4, no SATA −14.7, SATA whole −25.7, SATA per sub-band −25.7, oracle −31.3 dB. Bottom row: 2-D IRF of the representative target after SATA per sub-band and the spectral phase error along f_r and f_a.

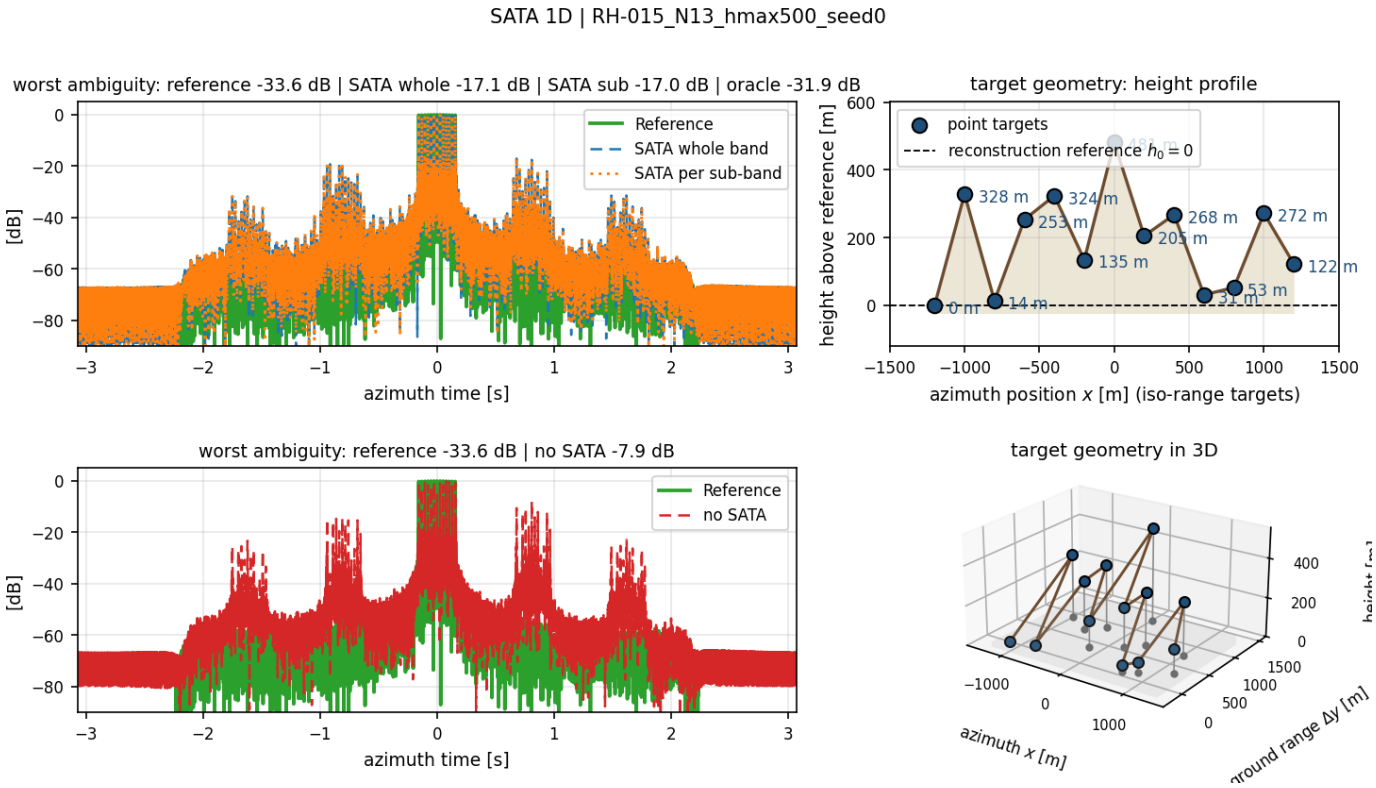


Figure 36: SATA 1D — Random heights, case `N13_hmax500_seed0`: $N = 13$ iso-range targets at x = -1200..1200 m (200 m apart), heights 0, 328, 14, 253, 324, 135, 481, 205, 268, 31, 53, 272, 122 m. Worst ambiguity: reference −33.6, no SATA −7.9, SATA whole −17.1, SATA per sub-band −17.0, oracle −31.9 dB.

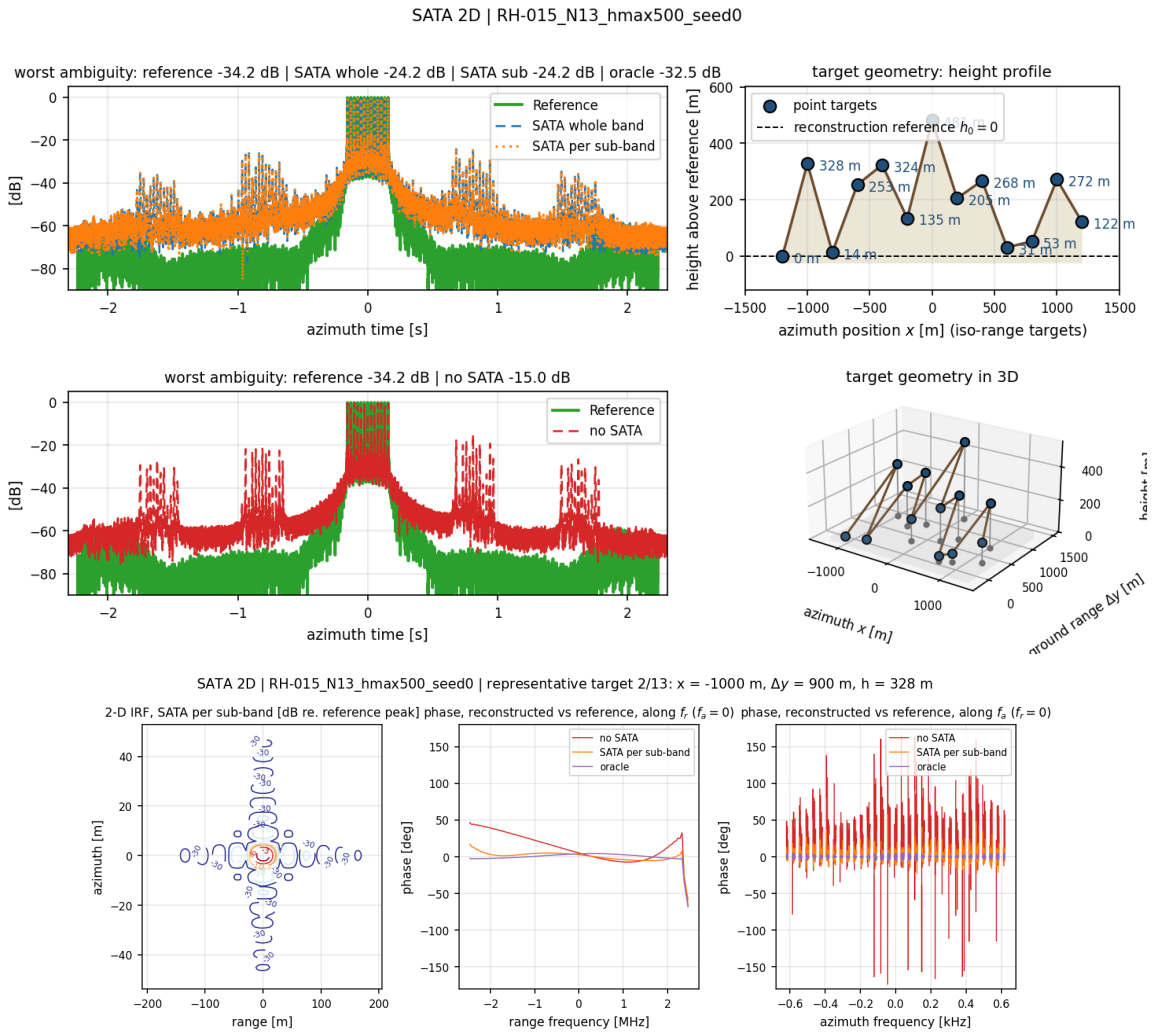


Figure 37: SATA 2D — Random heights, case `N13_hmax500_seed0`: $N = 13$ iso-range targets at x = -1200..1200 m (200 m apart), heights 0, 328, 14, 253, 324, 135, 481, 205, 268, 31, 53, 272, 122 m. Worst ambiguity: reference −34.2, no SATA −15.0, SATA whole −24.2, SATA per sub-band −24.2, oracle −32.5 dB. Bottom row: 2-D IRF of the representative target after SATA per sub-band and the spectral phase error along f_r and f_a.

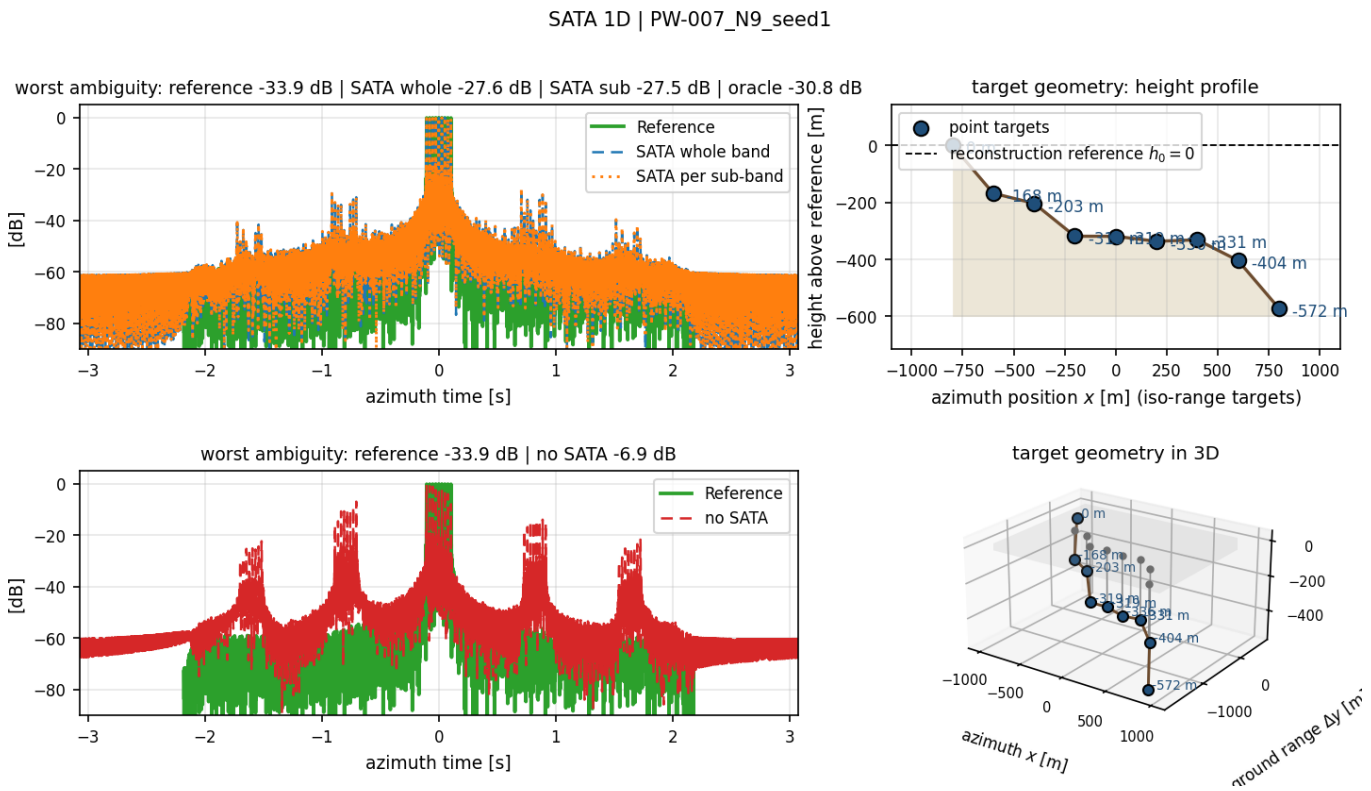


Figure 38: SATA 1D — Piecewise ramp, case `N9_seed1`: $N = 9$ iso-range targets at x = -800..800 m (200 m apart), heights 0, -168, -203, -319, -319, -336, -331, -404, -572 m. Worst ambiguity: reference −33.9, no SATA −6.9, SATA whole −27.6, SATA per sub-band −27.5, oracle −30.8 dB.

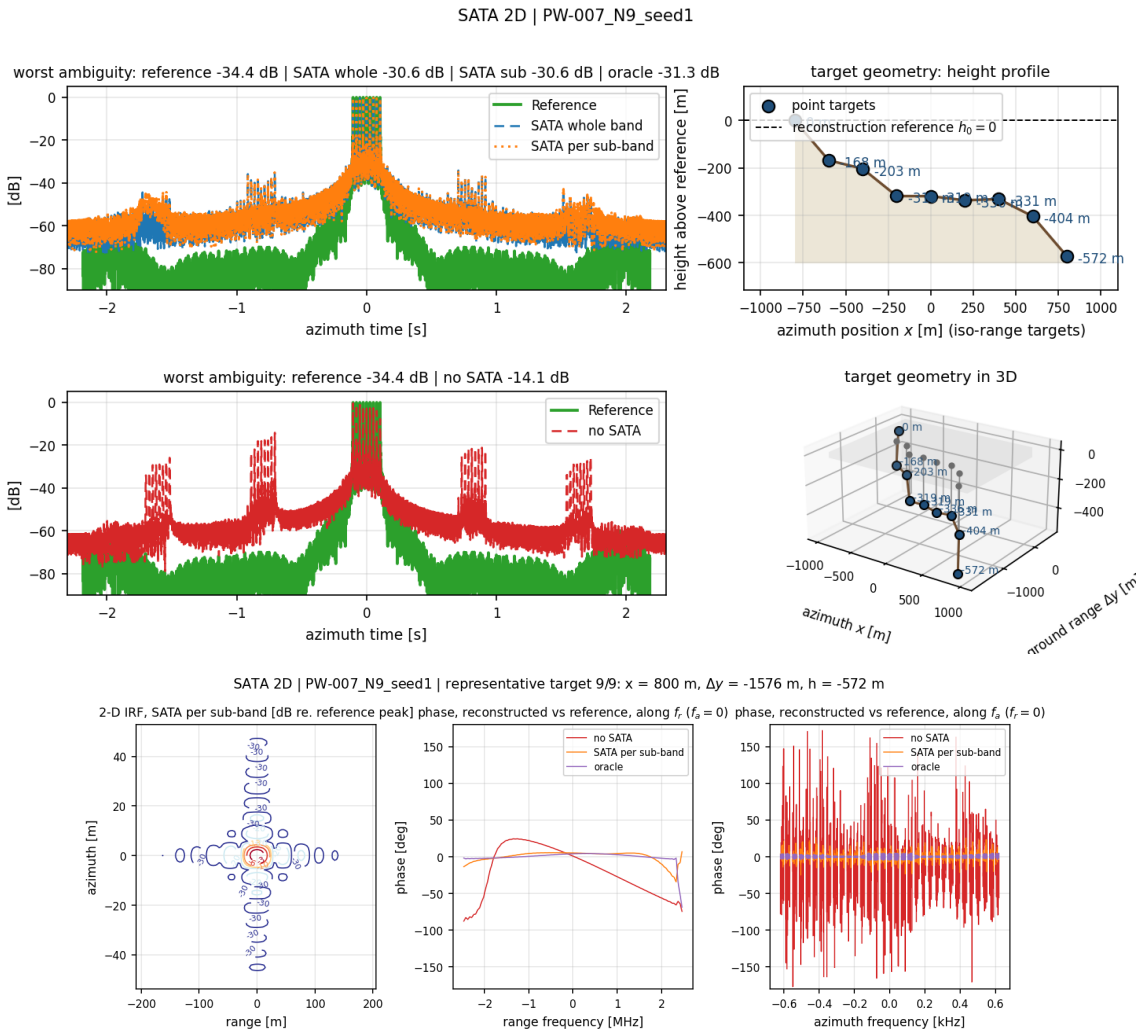


Figure 39: SATA 2D — Piecewise ramp, case `N9_seed1`: $N = 9$ iso-range targets at x = -800..800 m (200 m apart), heights 0, -168, -203, -319, -319, -336, -331, -404, -572 m. Worst ambiguity: reference −34.4, no SATA −14.1, SATA whole −30.6, SATA per sub-band −30.6, oracle −31.3 dB. Bottom row: 2-D IRF of the representative target after SATA per sub-band and the spectral phase error along f_r and f_a.

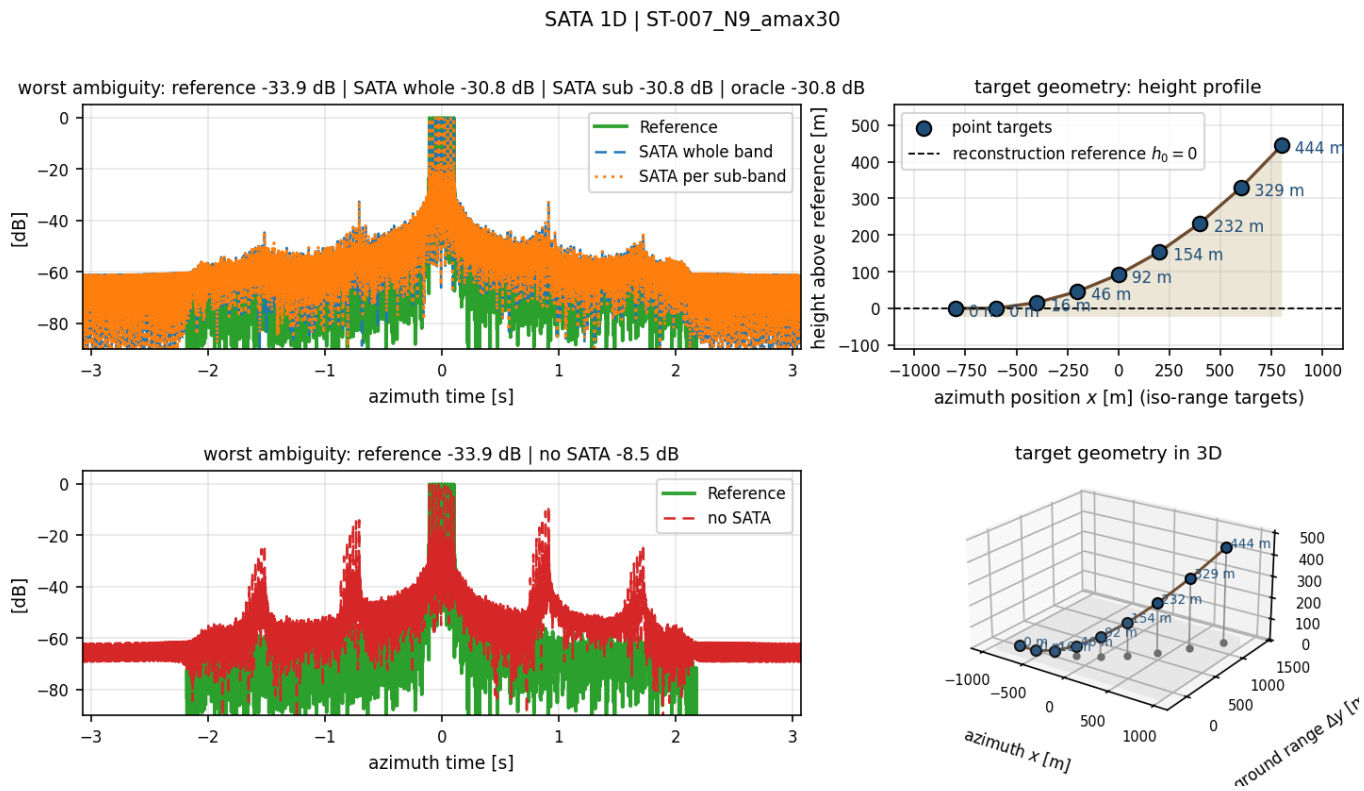


Figure 40: SATA 1D — Steepening ramp, case `N9_amax30`: $N = 9$ iso-range targets at x = -800..800 m (200 m apart), heights 0, 0, 16, 46, 92, 154, 232, 329, 444 m. Worst ambiguity: reference −33.9, no SATA −8.5, SATA whole −30.8, SATA per sub-band −30.8, oracle −30.8 dB.

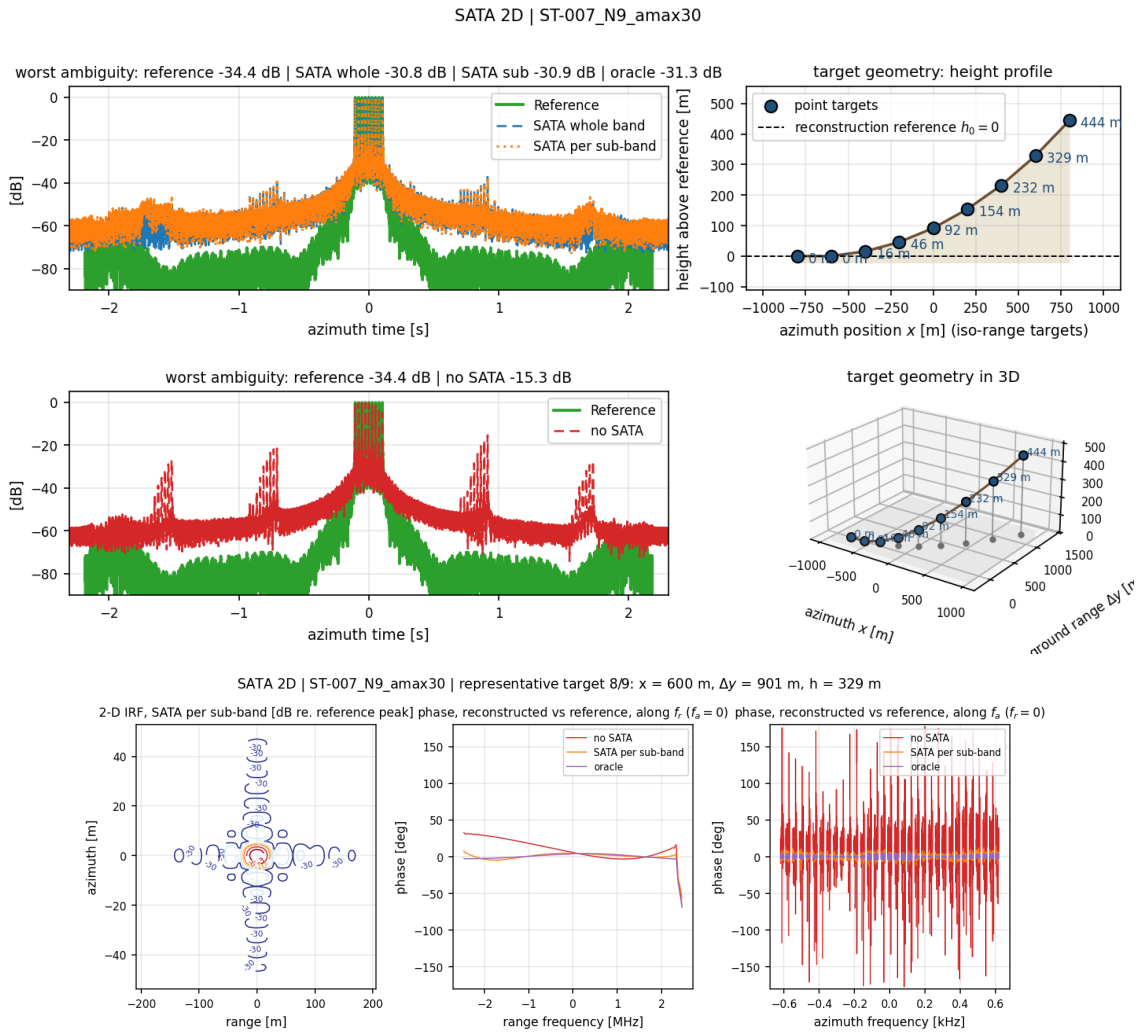


Figure 41: SATA 2D — Steepening ramp, case `N9_amax30`: $N = 9$ iso-range targets at x = -800..800 m (200 m apart), heights 0, 0, 16, 46, 92, 154, 232, 329, 444 m. Worst ambiguity: reference −34.4, no SATA −15.3, SATA whole −30.8, SATA per sub-band −30.9, oracle −31.3 dB. Bottom row: 2-D IRF of the representative target after SATA per sub-band and the spectral phase error along f_r and f_a.

10 Conclusions

- Against the reference codes, both implementations reproduce the sub-aperture sizing, the Doppler axis, the ruler and the rotation; the bookkeeping of the overlap is a different but equivalent scheme (tested: at the oracle where the reference cross-fade is 2 dB behind, and within 1–3 dB in the scenes beyond the resolution of the method); the truncation and the Nyquist label of the reference are replaced by rounding and by the wrapped axis without measurable effect.
- The map is the point-scene form of the reference's DEM lookup with what the sub-sampled channel line adds: the fold images. `sata2d` now also reproduces the range side of the reference lookup (`/norcmc`): the map read by a Doppler bin is that of the range the energy of that bin comes from. This is neutral for isolated scatterers and closes most of the remaining gap to the oracle on the cross-track ramps with scatterers one or two range bins apart.
- The terms of the reference that are not implemented (λ across the range band, residual range shifts) are below 2° or absent in this geometry.
- In the sweeps SATA 1D and SATA 2D give the same answer wherever the targets share a range line, which is every family except the cross-track ramp: the 2-D chain is the 1-D kernel per range line. Where SATA is within its resolution (ramps of moderate slope, cliffs of moderate jump, the along-track families up to about 100 m between neighbours 200 m apart) it reaches the oracle in both; where two scatterers closer than $\delta x = 309$ m differ by hundreds of metres (spike, mesa, random heights, steep ramps with many targets) both stop 7–14 dB short of the oracle, the resolution of the sub-aperture, not the implementation.
- The cross-track ramp separates the two: in 1D all its targets share one cell of one line and receive one correction (gain median 1 dB, erratic in sign, SATA −9.7 dB against an oracle at −38.9); in 2D each sits in its own range bins and SATA reaches −38.3 dB (median) against an oracle at −39.3 and a reference at −39.2, a gain of 12 dB (up to 26), monotonic in the slope — except at $\alpha = \theta$, where the ramp is iso-range and the 1D situation returns, and with a 4–7 dB gap where scatterers at the same azimuth are one or two range bins apart.
- The floor that the first 2-D sweep showed for flat targets away from the reference range was neither SATA's nor the reconstruction's: the ported signal generator samples a 5.7-sample chirp with hard edges, so channels with a sub-sample bistatic delay were not delayed copies of the reference. With the band-limited generator the chain reconstructs a flat target at the reference level at every range, and the 2-D results of this document are free of it; the earlier 2-D studies are not.

A Files and reproduction

sata1d_code/sata1d/sata.py	kernel (sata_kernel, option scheme), footprint map, sata_channels
sata1d_code/sata1d/subband.py	sub-band axis, residual per band, reconstruct_subband
sata1d_code/verify_sata1d.py	T1–T19
sata1d_code/run_geometry_sweep.py	the 1D sweep (figures with 2D and 3D geometry, .npz per case)
sata1d_code/probe_kernel_schemes.py	Table 3
sata2d/sata.py	kernel (_run_kernel with map stack and row_select, option scheme)
sata2d/reconstruction.py	ResidualMaps (get, rows, group_of), channels_subband_3d(rcm_lookup), reconstruct_subband_2d, azimuth_envelope, worst_ambiguity
sata2d/verify_sata2d.py	V1–V14
sata2d/run_geometry_sweep_2d.py	the 2D sweep (figures with the Fig. 4.20 panels, .npz per case)
sata2d/plot_sweep_overview.py, plot_case_pages.py, plot_sweep_1d_vs_2d.py	the figures of this document
sata2d/sweep_geometry_plots.py	the geometry panels (2D, 3D), shared

```
cd dlr-project/sata1d_code
python verify_sata1d.py
python run_geometry_sweep.py --bxt 100 --out geometry_sweep

cd dlr-project/sar_reconstruction
python -m sata2d.verify_sata2d
python -m sata2d.run_geometry_sweep_2d --out geometry_sweep_2d
python -m sata2d.plot_sweep_overview geometry_sweep_2d/sweep_results.csv --label "SATA
    2D" --tag 2d
python -m sata2d.plot_case_pages geometry_sweep_2d/bxt100/topo_ramp/TR-060_N9_a10.npz
    --out pages
```